

# PRAMAN AI

## Packaging Regulations & Automated Metrology Audit Network

*"From Package Image to Explainable Compliance Decision"*

| DOCUMENT CLASSIFICATION | TECHNICAL EVALUATION DOSSIER                                                        |
|-------------------------|-------------------------------------------------------------------------------------|
| Target Event / Review:  | Smart India Hackathon (SIH 2026) / Statutory Evaluator Review                       |
| Statutory Ground Truth: | Legal Metrology Act, 2009 & Packaged Commodities Rules, 2011 (40 Official Gazettes) |
| System Core:            | Computer Vision (OpenCV) + Tesseract OCR + Deterministic Legal Rule Engine          |
| Report Capabilities:    | Official Court-Ready PDF (ReportLab) & Editable Word Notice (DOCX)                  |
| Security & Roles:       | RBAC (ADMIN, SUPERVISOR, INSPECTOR) + SHA-256 JWT + Immutable Audit Logs            |
| Accessibility:          | Senior Mode, High Contrast Mode, Speech Synthesis Audio Guidance                    |
| Audited Codebase:       | 100% Verified FastAPI Backend + React 18 / Tailwind Frontend + SQLite DB            |
| Status:                 | Fully Functional Prototype with Live Pipeline & Curated Statutory Benchmarks        |

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### CONFIDENTIAL & AUTHORITATIVE PREPARATION GUIDE

This dossier contains the complete architectural, algorithmic, regulatory, and defensive knowledge base for PRAMAN AI. All data, logic, rules, and evaluator Q&As; match the actual verified implementation of the system.

TABLE OF CONTENTS & SITEMAP

| Part    | Dossier Section                                     | Key Scope & Content                                                                   |
|---------|-----------------------------------------------------|---------------------------------------------------------------------------------------|
| Part 1  | Understand the Entire Project & Verification Matrix | Implementation audit distinguishing Dynamic, Rule-based, AI, and Mock items           |
| Part 2  | Project Executive Summary & Pitches                 | 30-sec, 1-min, 2-min, and 5-min presentation-ready explanations                       |
| Part 3  | Problem Statement & Solution Matrix                 | Deep analysis of manual packaging inspection vs PRAMAN AI automation                  |
| Part 4  | Complete System Workflow                            | End-to-end 15-step execution pipeline from login to report generation                 |
| Part 5  | Professional Workflow Diagram                       | Visual flowchart of data transformations and inspector touchpoints                    |
| Part 6  | System Architecture & Diagram                       | Multi-tier architecture breakdown: Frontend, API, AI/OCR, Engine, DB, Reports         |
| Part 7  | Every Major Feature (1–24)                          | Deep-dive into all 24 UI/Backend modules with inputs, outputs, logic, and limitations |
| Part 8  | AI / Computer Vision / OCR Deep Dive                | OpenCV CLAHE, Tesseract 5.4, character confidence, and edge-case handling             |
| Part 9  | Legal Metrology / Compliance Logic                  | Detailed breakdown of the 12 Statutory Rule Groups indexed from 40 Gazettes           |
| Part 10 | Nutritional Values & Health Classification          | Nutrition facts extraction, FSSAI/ICMR health indicator, and legal distinction        |
| Part 11 | Visual Evidence & Bounding Boxes                    | Image-to-rule traceability, color overlays (Pass/Warn/Violate), crop cards            |
| Part 12 | Report Generation Engine                            | Official Court-Ready PDF (ReportLab) & Editable DOCX (python-docx) generators         |
| Part 13 | User Roles & Permissions (RBAC)                     | Role definitions (Admin, Supervisor, Inspector), badge numbers, access limits         |
| Part 14 | Complete Data Flow                                  | Data models, state transitions, JSON payloads, and storage schemas                    |
| Part 15 | Technology Stack & Tool Justifications              | FastAPI, React 18, OpenCV, Tesseract, ReportLab, SQLite/PostgreSQL                    |
| Part 16 | Technical Design Decisions                          | Architectural trade-offs: Deterministic Rules vs LLM, Local OCR vs Cloud APIs         |
| Part 17 | Brutally Honest Limitations                         | Current edge cases, image skew limits, multi-surface packages, regional fonts         |
| Part 18 | Security Architecture                               | Authentication, password hashing, session tokens, audit trail, production roadmap     |
| Part 19 | Scalability Roadmap                                 | Horizontal scaling, Celery queues, PostgreSQL sharding, cloud OCR clusters            |
| Part 20 | Real-World Government Deployment                    | NIC/e-Governance integration, handheld enforcement devices, offline edge sync         |
| Part 21 | Hackathon Evaluator Questions (A–AL)                | Exhaustive category-by-category defense questions with strong technical answers       |
| Part 22 | Hard Technical Questions & Traps                    | Adversarial questions, OCR failures, legal conflict resolution, mathematical proofs   |
| Part 23 | Demo Questions & Live Interruptions                 | Questions asked during live demonstration and step-by-step inspector responses        |
| Part 24 | Perfect Demo Walkthrough Script                     | Click-by-click, say-by-say presentation script for winning the evaluation             |
| Part 25 | Elevator Pitches (5 Variants)                       | 30s, 1m, 2m, Problem-first, and Innovation-first pitch scripts                        |
| Part 26 | Explain It to a Professor                           | Systems engineering, computer vision, deterministic logic, and software design review |
| Part 27 | Quick Revision Cheat Sheet                          | Summary cards, key statistics, penalty sections, and statutory definitions            |
| Part 28 | Questions We Must Be Able to Answer                 | Internal team readiness checklist before facing the judging panel                     |

PART 1 — UNDERSTAND THE ENTIRE PROJECT & VERIFICATION MATRIX

Ground-truth technical status of all system components verified against the codebase.

To ensure 100% academic integrity and factual defensibility during evaluation, every feature in PRAMAN AI has been audited to determine its exact architectural nature. Nothing is fabricated or claimed beyond actual code reality.

| Component / Feature            | Implementation Nature                              | Backend / Engine Driver                                    | Verification Status                              |
|--------------------------------|----------------------------------------------------|------------------------------------------------------------|--------------------------------------------------|
| Image Upload & Samples         | Actually Implemented (Dynamic + Curated Samples)   | FastAPI /api/scan/upload + disk storage                    | 100% Verified in scan_routes.py                  |
| Image Preprocessing            | Actually Implemented (Computer Vision)             | OpenCV CLAHE, bilateralFilter, Hough skew detection        | 100% Verified in preprocessor.py                 |
| OCR & Word Bounding Boxes      | Actually Implemented (AI/OCR Engine)               | Tesseract 5.4.0 via pytesseract.image_to_data              | 100% Verified in ocr_engine.py                   |
| Declaration Extraction         | Actually Implemented (Deterministic NLP/Regex)     | 9 specialized regex parsers with line grouping             | 100% Verified in declaration_extractor.py        |
| Nutritional Information        | Actually Implemented (Nutrient Parser)             | Parses 11 nutrient fields (Energy, Protein, Sugars, etc.)  | 100% Verified in declaration_extractor.py        |
| Health Classification          | Actually Implemented (Rule-Based Classifier)       | FSSAI/ICMR threshold evaluation (Sodium, Fat, Sugar)       | 100% Verified in nutrition_classifier.py         |
| Compliance Rule Engine         | Actually Implemented (Deterministic Rules)         | 12 Statutory Rule Groups from 40 Official Gazettes         | 100% Verified in engine.py & rule_definitions.py |
| Explainable Scoring (0-100)    | Actually Implemented (Weighted Scorer)             | 5 weighted categories (Completeness 40%, Units 25%, etc.)  | 100% Verified in scoring.py                      |
| Visual Evidence Bounding Boxes | Actually Implemented (Computer Vision)             | OpenCV rectangle rendering + UUID crop generation          | 100% Verified in visual_evidence.py              |
| Official PDF Report            | Actually Implemented (Document Engine)             | ReportLab Platypus PDF with GoI banner & signature box     | 100% Verified in pdf_report.py                   |
| Editable DOCX Report           | Actually Implemented (Document Engine)             | python-docx generator with formatted statutory notice      | 100% Verified in docx_report.py                  |
| Authentication & RBAC          | Actually Implemented (JWT + Salted SHA-256)        | 6 authorized official users (Admin, Supervisor, Inspector) | 100% Verified in auth.py & auth_routes.py        |
| Remembered Login & Fast-Switch | Actually Implemented (Frontend + Storage)          | LocalStorage persistence + 1-click role switcher           | 100% Verified in AuthContext.jsx                 |
| Accessibility (Senior/Speech)  | Actually Implemented (Browser Web APIs)            | Web Speech Synthesis API + CSS high-contrast filters       | 100% Verified in AccessibilityContext.jsx        |
| Immutable Audit Logging        | Actually Implemented (Database-Backed)             | SQLAlchemy AuditLog recording all inspections/logins       | 100% Verified in models.py & report_routes.py    |
| 40 Gazette Rule Library        | Actually Implemented (Dataset Index)               | 40 PDF gazettes in Government_Gazette_Dataset              | 100% Verified on filesystem & UI                 |
| Live Camera Capture            | Partially Implemented / File-upload primary        | Frontend file picker supports direct image file upload     | Live WebRTC stream is planned for mobile build   |
| Regional OCR (Hindi/Tamil)     | Partially Implemented (Tesseract multi-lang ready) | English dataset primary; Hindi OCR trained in Tesseract    | Indic-OCR pipeline documented in Scalability     |

PART 2 — PROJECT EXECUTIVE SUMMARY & PITCHES

High-impact summaries tailored for 30-second to 5-minute evaluator evaluations.

What is PRAMAN AI?

PRAMAN AI (*Packaging Regulations & Automated Metrology Audit Network*) is an AI-powered Legal Metrology compliance inspection platform designed specifically for government enforcement officers, legal metrology inspectors, and regional supervisors in India.

What Problem Does it Solve?

Pre-packaged commodities sold across India must strictly declare mandatory consumer safeguards under the Legal Metrology Act, 2009 and Packaged Commodities Rules (PCR), 2011. Manual inspection in wholesale markets and retail stores is slow (15–20 minutes per package), error-prone, highly subjective, lacks evidence preservation, and overwhelms field officers. PRAMAN AI automates the entire inspection in under 3 seconds.

**Who are the Intended Users?**

1. Field Enforcement Inspectors (on-site store audits), 2. Review Supervisors (approvals and notice issuance), 3. Central Ministry Administrators (state-wide compliance analytics and policy oversight).

**What is the Main Innovation?**

Unlike generic black-box vision models or hallucination-prone Large Language Models, PRAMAN AI pairs **Computer Vision OCR** with a **Deterministic Legal Rule Engine** derived directly from 40 Official Government Gazettes. Every pass or violation is mathematically proven, tied to exact statutory clauses (e.g., Rule 6(1)(c), GSR 226(E)), and accompanied by cropped visual evidence snippets.

## 30-Second Elevator Pitch

PRAMAN AI transforms Legal Metrology enforcement in India from a slow, manual 20-minute paperwork process into a 3-second automated audit. An enforcement officer snaps a photo of any packaged commodity; our Computer Vision and Deterministic Rule Engine extracts declarations, checks compliance against 40 Official Gazettes, flags violations with visual bounding boxes, and generates court-ready statutory inspection notices in PDF and editable Word formats.

## 1-Minute Standard Pitch

In India, over 100 million pre-packaged commodities are sold daily, yet Legal Metrology enforcement relies on manual inspections. Officers struggle with complex amendments like Unit Sale Price rules, metric quantity standards, and tax clauses. PRAMAN AI solves this by automating label compliance. When an image is uploaded, OpenCV enhances the label, Tesseract extracts text bounding boxes, and our deterministic rule engine evaluates 12 statutory rule groups derived from official gazettes. It scores the package from 0 to 100, detects infractions such as misleading 'Jumbo' quantities or missing helpline emails, crops visual evidence cards, and exports official statutory notices instantly. It also includes senior-friendly accessibility and immutable audit logging.

## 2-Minute Technical Pitch

PRAMAN AI is architected with a decoupled FastAPI backend and React frontend. The pipeline begins with OpenCV preprocessing—applying CLAHE for uneven lighting, bilateral filtering for noise reduction, and Hough transform skew correction. Tesseract 5.4 extracts word-level bounding boxes and confidence scores. Next, our deterministic declaration extractor parses mandatory fields—Commodity Name, Manufacturer Address with Pincode, Metric Net Quantity, MRP with tax clauses, Unit Sale Price, and Consumer Care details. In parallel, it extracts 11 nutritional facts and derives an informational health classification. The compliance engine evaluates 12 statutory rule groups derived from 40 indexed gazette PDFs, computing a weighted 0–100 score across 5 categories. High-visibility bounding boxes are rendered directly on the package, and ReportLab and python-docx engines generate official notices with digital signatures and audit logs.

## 5-Minute Technical Deep-Dive

PRAMAN AI addresses the core bottleneck of statutory enforcement: the need for absolute legal determinism without AI hallucination. Large Language Models cannot be used for court prosecution because their non-deterministic nature introduces hallucinations. PRAMAN AI deliberately uses a hybrid pipeline: AI for Computer Vision and OCR (perceptual task), coupled with a deterministic statutory rule engine (symbolic reasoning). The architecture uses SQLite/PostgreSQL with SQLAlchemy ORM, FastAPI REST endpoints, and JWT authentication with Role-Based Access Control. If an image has poor contrast, OpenCV calculates brightness and contrast standard deviations, dynamically triggering legibility warnings under Rule 7 and 9. The system validates unit conversions—ensuring USP equals MRP divided by net quantity per GSR 226(E). If a product is non-compliant, Section 36(1) show-cause notice recommendations are generated automatically, complete with high-resolution evidence snippets.

PART 3 — PROBLEM STATEMENT & SOLUTION MATRIX

Comprehensive comparison between manual packaging inspection and PRAMAN AI automation.

The Legal Metrology (Packaged Commodities) Rules, 2011 mandate at least 9 distinct declarations on every consumer package. In practice, manual enforcement suffers from severe systemic limitations:

| Manual Inspection Problem       | Root Cause & Operational Impact                                                                                                                           | PRAMAN AI Automated Solution                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Extreme Time Consumption        | Takes 15–20 minutes per package to verify 9 declarations, cross-reference gazettes, and hand-write inspection memos.                                      | <b>3-Second End-to-End Audit:</b> Instant OpenCV/Tesseract extraction and automated rule evaluation.                             |
| Human Error & Fatigue           | Inspectors easily miss micro-lettering, missing 'inclusive of all taxes' clauses, or subtle non-metric units ('gms' vs 'g').                              | <b>100% Deterministic Verification:</b> Regex and NLP algorithms strictly validate metric symbols and statutory tax clauses.     |
| Regulatory Complexity           | Over 40 gazette amendments exist (e.g., GSR 226(E) for Unit Sale Price, 2022 Garment Rules, 2023 QR Code proviso). Officers cannot recall all amendments. | <b>Automated Gazette Ground-Truth:</b> 12 rule groups continuously reflect the 40 indexed gazette PDFs.                          |
| Lack of Visual Evidence         | Officers take manual mobile photos that lack coordinate mapping, making evidence contestable in consumer courts.                                          | <b>Color-Coded Bounding Overlays:</b> Auto-generated high-resolution cropped evidence cards tied to exact violation coordinates. |
| Subjective Compliance Decisions | Different inspectors apply different penalties for identical infractions, leading to accusations of bias.                                                 | <b>Standardized 0–100 Score:</b> Explainable, weighted scoring across 5 distinct regulatory categories.                          |
| Paperwork & Notice Delays       | Typing formal Show-Cause notices under Section 36(1) takes days, delaying legal prosecution.                                                              | <b>1-Click Official Reports:</b> Instant court-ready PDF and editable Word (DOCX) notice generation.                             |
| Audit Trail & Tampering         | Physical inspection logs are susceptible to loss, unauthorized changes, or lack of supervisory oversight.                                                 | <b>Immutable Audit Logging:</b> Every scan, review, and supervisor note is timestamped and recorded in SQL.                      |

PART 4 — COMPLETE SYSTEM WORKFLOW

Step-by-step execution pipeline from authentication to report export.

Step 1: Secure Enforcement Login

Officer authenticates using badge credentials or one-click role switcher. System validates JWT access token and assigns role permissions (ADMIN, SUPERVISOR, INSPECTOR). An immutable login event is appended to the Audit Trail.

Step 2: Executive Dashboard Overview

Officer reviews state-wide compliance statistics, compliance rate gauge (e.g. 66.7%), top violations breakdown, category distribution, and recent inspection activity feed.

Step 3: Scan Packaging Studio Initiation

Officer navigates to Scan Studio. Selects either direct camera/file image upload or loads one of the 3 SIH curated statutory demo packages (Atta - Compliant, Sunflower Oil - Non-Compliant, Namkeen - Critical Violation).

Step 4: Image Ingestion & Disk Storage

FastAPI backend ingests the image file, assigns a unique Inspection ID (`PRM-YYYYMMDD-XXXXXX`), and stores the raw image in the backend upload repository.

Step 5: OpenCV Computer Vision Preprocessing

ImagePreprocessor loads the image matrix. Converts to grayscale, calculates mean brightness and contrast STD. Applies CLAHE (clipLimit=2.0) for adaptive contrast equalization, bilateral denoising to preserve crisp text edges, and Canny/Hough transform skew correction.

Step 6: Tesseract 5.4 OCR & Bounding Box Extraction

OCREngine runs Tesseract with `image\_to\_data`, extracting word-level text, coordinates (x, y, w, h), line groupings, block numbers, and character confidence scores. Generates aggregated lines with spatial coordinates.

Step 7: Mandatory Declaration Extraction (Deterministic NLP)

DeclarationExtractor runs 9 specialized regex and linguistic parsers across the OCR lines to extract: Commodity Name, Manufacturer/Packer Address with Pincode, Metric Net Quantity, Manufacturing Date, MRP with Tax Clause, Unit Sale Price (USP), Consumer Care Helpline/Email, Country of Origin, and Garment Sizes.

**Step 8: Nutritional Facts & Informational Health Assessment**

Nutrient parser extracts Energy, Protein, Carbohydrates, Sugars, Added Sugars, Total Fat, Saturated Fat, Trans Fat, Sodium/Salt, and Dietary Fiber. NutritionClassifier evaluates values against FSSAI/ICMR public health thresholds to produce an informational health indicator (HEALTHIER, MODERATE, or LESS HEALTHY).

**Step 9: Deterministic Legal Metrology Rule Engine**

ComplianceRuleEngine executes 12 statutory rule evaluations derived from the 40 indexed gazette PDFs. Evaluates address completeness, metric unit legality (Rule 11-13), tax clause presence (Rule 6(1)(e)), mathematical USP accuracy (GSR 226(E)), and QR code proviso compliance.

**Step 10: Explainable Compliance Scoring (0–100)**

ComplianceScorer computes weighted scores across 5 categories: Declaration Completeness (40%), Value & Unit Validation (25%), Price & USP Consistency (15%), Consumer Redressal & Origin (10%), and Display Legibility (10%). Formulates regulatory status (COMPLIANT, NON-COMPLIANT, PENDING REVIEW) and statutory recommendations under Section 36(1).

**Step 11: Visual Evidence & Overlay Generation**

VisualEvidenceGenerator draws color-coded bounding boxes on the packaging image: Green (PASS), Amber (WARNING), Red (VIOLATION). Crops localized high-resolution evidence snippets for each infraction and saves annotated images.

**Step 12: Database Persistence**

SQLAlchemy session creates or updates the Product entity, creates the primary Inspection record, and inserts individual records into ExtractedDeclarations, ComplianceResults, Violations, and AuditLogs.

**Step 13: Interactive UI Audit Studio**

Frontend displays the Score Gauge, Interactive Visual Evidence Viewer, Declarations Audit table with confidence bars, Nutritional Profile card, and Rule Citations.

**Step 14: Supervisory Review & Action Workflow**

Supervisor or Admin can add supervisory notes and update the official inspection state (COMPLETED, APPROVED, UNDER\_REVIEW, FLAGGED).

**Step 15: Official PDF & DOCX Export**

Officer clicks 'Download Official PDF' or 'Export DOCX'. ReportLab / python-docx dynamically generates official Government of India formatted notices complete with metadata, violation tables, and signature endorsement blocks.

PART 5 — PROFESSIONAL WORKFLOW DIAGRAM

Visual flowchart of PRAMAN AI end-to-end data pipeline and inspector touchpoints.

The following structural diagram visualizes the end-to-end PRAMAN AI pipeline, highlighting data transformations and human-in-the-loop inspector decision points:

| PIPELINE STAGE            | TRANSFORMATION & ARTIFACTS                                                                                         | ACTOR / INTERACTION                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1. INGESTION              | Package Image Upload / Demo Scenario Selection → Disk Ingestion & ID Allocation (PRM-...)                          | Enforcement Inspector (Field / Studio)                   |
| 2. COMPUTER VISION        | OpenCV CLAHE Histogram Equalization + Bilateral Filtering + Hough Transform Deskewing                              | Automated Pipeline (preprocessor.py)                     |
| 3. AI OCR ENGINE          | Tesseract 5.4.0 Engine → Raw Text Stream + Word Bounding Box Coordinates + Character Confidence                    | Automated Pipeline (ocr_engine.py)                       |
| 4. INFORMATION EXTRACTION | Deterministic NLP & Regex Parsers → Structured Declarations (Mfr, Qty, MRP, USP, Date, Helpline) + Nutrition Facts | Automated Pipeline (declaration_extractor.py)            |
| 5. RULE VALIDATION        | Deterministic Legal Engine (12 Rules from 40 Gazettes) + FSSAI/ICMR Health Classification                          | Automated Pipeline (engine.py & nutrition_classifier.py) |
| 6. COMPLIANCE DECISION    | Explainable Weighted Scoring (0–100) → Status Decision (COMPLIANT, NON-COMPLIANT, PENDING REVIEW)                  | Automated Pipeline (scoring.py)                          |
| 7. VISUAL EVIDENCE        | OpenCV Bounding Box Overlay (Green/Amber/Red) + Cropped Evidence Snippet Card Generation                           | Automated Pipeline (visual_evidence.py)                  |
| 8. INSPECTION DETAIL UI   | Interactive Score Gauge, 6-Tab Audit Studio, Bounding Box Inspector, Audio Guidance                                | Inspector / Supervisor (Interactive Review)              |
| 9. SUPERVISORY WORKFLOW   | Supervisory Review Notes + Status Transition (APPROVED / FLAGGED / UNDER_REVIEW)                                   | Review Supervisor / Admin                                |
| 10. STATUTORY EXPORT      | ReportLab Court-Ready PDF Notice + python-docx Editable Word Notice + Immutable Audit Log                          | Enforcement Officer / Legal Wing                         |

PART 6 — SYSTEM ARCHITECTURE & DIAGRAM

Technical breakdown of frontend, API gateway, computer vision, rule engine, storage, and report layers.

PRAMAN AI follows a clean, decoupled, multi-tier architecture designed for high throughput, sub-second execution, and zero AI hallucination:

| Architectural Layer          | Technologies & Frameworks                                                | Functional Responsibility                                                                                                                       |
|------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Presentation Layer (UI)   | React 18, Vite, Tailwind CSS, Lucide Icons, Web Speech Synthesis API     | Responsive enforcement dashboard, interactive scan studio, 6-tab audit viewer, visual bounding box inspector, senior-friendly accessibility.    |
| 2. API Gateway & Routing     | FastAPI, Uvicorn ASGI Server, Pydantic Schemas, OAuth2 Password Bearer   | RESTful endpoints for auth, scanning, inspection history, product catalogue, rule library, analytics, and report downloads.                     |
| 3. Computer Vision & OCR     | OpenCV 4.10, Tesseract OCR 5.4.0, NumPy, Pillow                          | CLAHE contrast equalization, bilateral denoising, Hough deskewing, word-level bounding box coordinate extraction, character confidence scoring. |
| 4. Information Extraction    | Deterministic Regex Engine, Spatial Line Grouping, NLP Parsers           | Structured declaration parsing (Mfr, Net Qty, MRP, USP, Dates, Consumer Care) + 11 nutrient facts extraction.                                   |
| 5. Compliance & Scoring      | Deterministic Rule Engine, FSSAI/ICMR Health Classifier, Weighted Scorer | 12 Statutory Rule Groups from 40 Gazettes, explainable 0–100 scoring across 5 categories, show-cause notice recommendations.                    |
| 6. Visual Evidence Generator | OpenCV Graphic Overlay Engine, Bounding Box Cropper                      | Draws color-coded bounding overlays on packages and crops evidence snippets for reports and UI cards.                                           |
| 7. Statutory Report Engine   | ReportLab 4.5 Platypus (PDF) & python-docx (DOCX)                        | Generates court-admissible PDF notices with GoI headers and editable Word documents for legal proceedings.                                      |
| 8. Database & Persistence    | SQLAlchemy ORM, SQLite (Built-in) / PostgreSQL Ready                     | Relational storage for Users, Products, Inspections, Declarations, ComplianceResults, Violations, Reports, and AuditLogs.                       |
| 9. Security & Audit Trail    | PyJWT, SHA-256 + Salted Hashing, RBAC Middleware                         | Role-Based Access Control (Admin, Supervisor, Inspector), session validation, immutable audit logging of all system actions.                    |

## PART 7 — EVERY MAJOR FEATURE DEEP-DIVE (FEATURES 1–24)

Exhaustive documentation of all 24 UI and backend modules with inputs, outputs, logic, and limitations.

### 1. Officer Authentication & Demo Login

**Core Function:** Authenticates enforcement personnel and establishes RBAC context.

|                      |                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer enters email/password or clicks a 1-click official user card.                                 |
| Input Data:          | User credentials (email, username, password).                                                         |
| Output Produced:     | JWT access token (HS256) + User Profile object + Role context.                                        |
| Internal Logic:      | Backend queries User table, computes salted SHA-256 hash, generates JWT, logs USER_LOGIN audit event. |
| Operational Utility: | Prevents unauthorized public access while allowing seamless field demonstration.                      |
| Underlying Tech:     | FastAPI OAuth2, PyJWT, SHA-256, LocalStorage.                                                         |
| Failure Handling:    | Incorrect password returns 401 Unauthorized; token expiry logs user out.                              |
| Limitations:         | Demo secret key in environment; production requires asymmetric RS256 and OAuth2 IDP.                  |

### 2. Executive Enforcement Dashboard

**Core Function:** Provides high-level situational awareness across all inspections.

|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| User Interaction:    | Officer views dashboard cards, compliance rate charts, top violations, and recent scans.      |
| Input Data:          | Database aggregation queries via /api/analytics/summary.                                      |
| Output Produced:     | Total scan count, compliant count, non-compliant count, compliance rate, violation chart.     |
| Internal Logic:      | Calculates SQL aggregate metrics (counts, percentages, top violation group-by queries).       |
| Operational Utility: | Enables supervisors to identify non-compliant market trends and offending manufacturers.      |
| Underlying Tech:     | React, Tailwind, SQLAlchemy func.count, Recharts / CSS Gauges.                                |
| Failure Handling:    | Empty database shows zero states.                                                             |
| Limitations:         | Currently single-node in-memory SQL aggregation; production requires OLAP / ClickHouse cache. |

### 3. Scan Packaging Studio

**Core Function:** Interactive studio for uploading package images or executing curated SIH demo scenarios.

|                      |                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer drags/drops package image or clicks one of 3 SIH curated demo sample cards.               |
| Input Data:          | Multipart/form-data image file OR sample filename string.                                         |
| Output Produced:     | Triggers 5-stage live AI inspection pipeline and transitions to Detail view.                      |
| Internal Logic:      | Handles file ingestion, creates unique Inspection ID, invokes OpenCV, OCR, and Compliance Engine. |
| Operational Utility: | Central workbench for field inspectors auditing pre-packaged goods.                               |
| Underlying Tech:     | FastAPI UploadFile, FileReader API, CSS Pulse progress indicators.                                |
| Failure Handling:    | Unsupported file formats (.bmp, .tiff) or corrupt images raise 400 Bad Request.                   |
| Limitations:         | Requires reasonable image resolution (>300 DPI recommended for micro-text).                       |

### 4. OCR & Text Detection Engine

**Core Function:** Converts package raster images into structured text tokens with spatial coordinates.

|                      |                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------|
| User Interaction:    | Automated background execution during scan.                                                         |
| Input Data:          | OpenCV preprocessed numpy image array.                                                              |
| Output Produced:     | Dictionary containing raw text, word bounding boxes, line groups, average confidence score.         |
| Internal Logic:      | Executes pytesseract.image_to_data, groups adjacent words into lines, computes line bounding boxes. |
| Operational Utility: | Extracts text from complex, unformatted packaging labels without human typing.                      |
| Underlying Tech:     | Tesseract OCR 5.4.0, PIL / Pillow, PyTesseract.                                                     |
| Failure Handling:    | Extremely blurry images yield low confidence (<50%) or empty text.                                  |
| Limitations:         | Tesseract CPU-bound; curved surface text requires cylindrical unrolling in pre-processing.          |

### 5. Information Extraction Engine

**Core Function:** Transforms unstructured OCR text into 9 mandatory Legal Metrology declarations.

|                      |                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| User Interaction:    | Automated execution following OCR.                                                                                                 |
| Input Data:          | OCR line blocks and raw text string.                                                                                               |
| Output Produced:     | Structured dictionary containing 9 declaration objects with detected values, boolean flags, confidence scores, and bounding boxes. |
| Internal Logic:      | Executes 9 deterministic regular expression parsers with contextual boundary cleaning.                                             |
| Operational Utility: | Bridges the gap between raw OCR text and statutory rule validation.                                                                |
| Underlying Tech:     | Deterministic Regex, Contextual String Tokenization.                                                                               |
| Failure Handling:    | Non-standard packaging terminology may fail pattern matching, triggering manual review.                                            |
| Limitations:         | Requires standard alphanumeric character sets; highly stylized artistic brand logos may not parse as plain text.                   |

6. Label Compliance Engine (12 Rules)

Core Function: Evaluates extracted declarations against 12 Statutory Legal Metrology Rule Groups.

|                      |                                                                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Interaction:    | Automated deterministic execution.                                                                                                                                             |
| Input Data:          | Extracted declarations dictionary + Image quality metrics.                                                                                                                     |
| Output Produced:     | Evaluation results list with PASS / WARNING / VIOLATION status, severities, and gazette citations.                                                                             |
| Internal Logic:      | Runs deterministic validation algorithms for each rule group (e.g. checks PIN code in address, SI unit symbols in Net Qty, inclusive of taxes in MRP, USP mathematical check). |
| Operational Utility: | Guarantees 100% legally grounded, explainable compliance decisions without AI hallucination.                                                                                   |
| Underlying Tech:     | Python Symbolic Rule Evaluator, Legal Metrology Act 2009 & PCR 2011 dataset.                                                                                                   |
| Failure Handling:    | If OCR misses a declaration entirely, rule engine records it as a VIOLATION.                                                                                                   |
| Limitations:         | Covers 12 major gazette rule groups; niche commodity exemptions require future rule expansion.                                                                                 |

7. Nutritional Values Extraction

Core Function: Extracts 11 statutory nutritional facts from the package Nutrition Facts panel.

|                      |                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| User Interaction:    | Automated execution during scan.                                                                                                         |
| Input Data:          | OCR text stream and line blocks.                                                                                                         |
| Output Produced:     | Nutritional facts dictionary (Energy, Protein, Carbs, Sugars, Added Sugars, Fiber, Fat, Saturated Fat, Trans Fat, Sodium, Serving Size). |
| Internal Logic:      | Regex parser detects nutrient keywords, numerical values, and unit tokens (kcal, g, mg). Standardizes Salt to Sodium equivalent.         |
| Operational Utility: | Enables dual-purpose consumer protection (metrology compliance + nutritional awareness).                                                 |
| Underlying Tech:     | Deterministic Regex Parser, Nutritional Unit Normalizer.                                                                                 |
| Failure Handling:    | Packages without nutritional tables return empty/not-found states gracefully.                                                            |
| Limitations:         | Extracts values per 100g/100ml; per-serving calculations depend on declared serving sizes.                                               |

8. Informational Health Indicator

Core Function: Provides an AI-derived public health classification based on detected nutrition facts.

|                      |                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User Interaction:    | Automated execution following nutritional extraction.                                                                                                        |
| Input Data:          | Extracted nutrition facts dictionary.                                                                                                                        |
| Output Produced:     | Health classification (HEALTHIER, MODERATE, LESS HEALTHY, or INSUFFICIENT DATA), score, risk factors, and positive factors.                                  |
| Internal Logic:      | Evaluates detected Sodium (>600mg), Saturated Fat (>5g), and Sugars (>22.5g) against standard FSSAI/ICMR benchmarks. Handles pure cooking oils contextually. |
| Operational Utility: | Offers transparent consumer health context without medical claims.                                                                                           |
| Underlying Tech:     | FSSAI & ICMR Public Health Dietary Benchmarks.                                                                                                               |
| Failure Handling:    | If nutrition panel is missing, outputs 'INSUFFICIENT DATA' rather than guessing.                                                                             |
| Limitations:         | Explicitly informational; does not constitute clinical dietary diagnosis.                                                                                    |

9. Violations Detection & Classification

Core Function: Aggregates and classifies infractions by severity level (CRITICAL, HIGH, MEDIUM, LOW).

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| User Interaction: | Displayed in the Violations tab of the Inspection Detail page. |
| Input Data:       | Compliance results from the rule engine.                       |

|                             |                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Output Produced:</b>     | List of Violation objects with titles, legal basis citations, severity badges, and evidence links.     |
| <b>Internal Logic:</b>      | Filters compliance results for VIOLATION and WARNING states; maps statutory legal basis from gazettes. |
| <b>Operational Utility:</b> | Provides enforcement officers with an immediate summary of actionable legal infractions.               |
| <b>Underlying Tech:</b>     | Relational Violation models, SQLAlchemy ORM.                                                           |
| <b>Failure Handling:</b>    | None (deterministic filter).                                                                           |
| <b>Limitations:</b>         | Severity ratings are calibrated to statutory penalties under Section 36(1) and PCR 2011.               |

## 10. Visual Evidence Bounding Boxes & Crops

**Core Function:** Renders color-coded overlays on the packaging image and crops violation snippets.

|                             |                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User Interaction:</b>    | Displayed in Evidence tab and on inspection cards.                                                                                                       |
| <b>Input Data:</b>          | Original packaging image + Compliance results with bounding box coordinates.                                                                             |
| <b>Output Produced:</b>     | Annotated packaging image URL + Array of cropped evidence JPEG files.                                                                                    |
| <b>Internal Logic:</b>      | OpenCV draws colored rectangles (Green=PASS, Amber=WARNING, Red=VIOLATION) with text badges. Crops localized bounding box coordinates with 20px padding. |
| <b>Operational Utility:</b> | Creates tamper-evident visual proof for prosecution in consumer dispute forums.                                                                          |
| <b>Underlying Tech:</b>     | OpenCV cv2.rectangle, cv2.putText, image array slicing.                                                                                                  |
| <b>Failure Handling:</b>    | Bounding boxes outside image boundaries are safely clamped.                                                                                              |
| <b>Limitations:</b>         | Coordinate accuracy depends on OCR word localization fidelity.                                                                                           |

## 11. Interactive Inspection Detail Studio

**Core Function:** Comprehensive 6-tab inspection workbench for deep-dive analysis.

|                             |                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>User Interaction:</b>    | Officer switches tabs (Label Compliance, Nutrition, Violations, Visual Evidence, Official Report, Supervisory Review). |
| <b>Input Data:</b>          | Inspection record ID.                                                                                                  |
| <b>Output Produced:</b>     | Interactive scorecard, score gauge, declaration audit table, nutrition grid, and zoomable evidence viewer.             |
| <b>Internal Logic:</b>      | Fetches complete inspection payload from /api/inspections/{id} and renders dynamic tabbed UI.                          |
| <b>Operational Utility:</b> | Consolidates all inspection artifacts into a single unified workspace.                                                 |
| <b>Underlying Tech:</b>     | React Tabs, Tailwind CSS, Lucide Icons, ScoreGauge component.                                                          |
| <b>Failure Handling:</b>    | Network failure displays friendly error card with return button.                                                       |
| <b>Limitations:</b>         | Optimized for desktop/tablet enforcement monitors.                                                                     |

## 12. Official PDF Report Generator

**Core Function:** Generates court-admissible statutory inspection reports in PDF format.

|                             |                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>User Interaction:</b>    | Officer clicks 'Download Official PDF'.                                                                              |
| <b>Input Data:</b>          | Full inspection payload dictionary.                                                                                  |
| <b>Output Produced:</b>     | High-resolution PDF file with Government of India header, metadata tables, declaration audits, and signature blocks. |
| <b>Internal Logic:</b>      | ReportLab Platypus compiles Flowables, Paragraphs, Tables, and HR lines into a structured letter-size PDF document.  |
| <b>Operational Utility:</b> | Serves as the official legal notice issued to non-compliant manufacturers.                                           |
| <b>Underlying Tech:</b>     | ReportLab 4.5.1 Platypus, Flowable Architecture.                                                                     |
| <b>Failure Handling:</b>    | Missing output directory is created automatically on demand.                                                         |
| <b>Limitations:</b>         | PDF is generated server-side; sub-second generation time (<200ms).                                                   |

## 13. Editable DOCX Report Export

**Core Function:** Generates fully editable Microsoft Word inspection notices for legal adaptation.

|                             |                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------|
| <b>User Interaction:</b>    | Officer clicks 'Export DOCX'.                                                                           |
| <b>Input Data:</b>          | Full inspection payload dictionary.                                                                     |
| <b>Output Produced:</b>     | Structured .docx document containing notice headings, findings tables, and violation summaries.         |
| <b>Internal Logic:</b>      | python-docx builds tables, headings, paragraph runs, and styling tags into a standard OpenXML document. |
| <b>Operational Utility:</b> | Allows legal officers to customize notice wording or add legal case numbers before dispatch.            |
| <b>Underlying Tech:</b>     | python-docx 1.1.2.                                                                                      |
| <b>Failure Handling:</b>    | None.                                                                                                   |

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Limitations: | Formatting conforms to standard GoI administrative notice layout. |
|--------------|-------------------------------------------------------------------|

## 14. Inspection History & Repository

**Core Function:** Searchable and filterable archive of all historical packaging inspections.

|                      |                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer searches by product name, ID, or filters by status (COMPLIANT, NON-COMPLIANT) and category. |
| Input Data:          | Filter query parameters (status, category, search text).                                            |
| Output Produced:     | List of historical inspection summary cards with thumbnail images, scores, and timestamps.          |
| Internal Logic:      | SQLAlchemy runs filtered ILIKE queries on Inspection table ordered by creation date descending.     |
| Operational Utility: | Enables repeat-offender tracking and audit trail review.                                            |
| Underlying Tech:     | FastAPI Query parameters, React Search/Filter bar.                                                  |
| Failure Handling:    | Empty search results display 'No matching inspections found'.                                       |
| Limitations:         | Pagination recommended for deployments exceeding 100,000 inspection records.                        |

## 15. Product Repository

**Core Function:** Centralized catalog of all scanned pre-packaged commodities.

|                      |                                                                                |
|----------------------|--------------------------------------------------------------------------------|
| User Interaction:    | Officer views product catalog, brands, declared MRPs, and net quantities.      |
| Input Data:          | Product database records.                                                      |
| Output Produced:     | Product cards with brand info, category badges, and linked inspection records. |
| Internal Logic:      | Relational Product queries linked via 1-to-many relationship with Inspections. |
| Operational Utility: | Provides brand-level compliance tracking across multiple SKUs.                 |
| Underlying Tech:     | SQLAlchemy Product model.                                                      |
| Failure Handling:    | Unbranded items default to 'Unbranded'.                                        |
| Limitations:         | Barcode scanning integration is pre-structured in the schema.                  |

## 16. Rule Library (40 Gazettes)

**Core Function:** Interactive knowledge base indexing the 40 official Legal Metrology gazettes and rule definitions.

|                      |                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer browses the 12 rule definitions, gazette citations, required declarations, and statutory penalties. |
| Input Data:          | Rule definitions array in rule_definitions.py.                                                              |
| Output Produced:     | Expandable rule cards with gazette filenames, rule numbers, severity tags, and explanatory notes.           |
| Internal Logic:      | Renders structured metadata mapped directly to the 40 PDFs in Government_Gazette_Dataset.                   |
| Operational Utility: | Ensures 100% legal transparency and serves as a training ground for junior inspectors.                      |
| Underlying Tech:     | React UI, Statutory Gazette Dataset Index.                                                                  |
| Failure Handling:    | None.                                                                                                       |
| Limitations:         | Reflects all 40 indexed gazettes in the SIH repository.                                                     |

## 17. Immutable Audit Trail & Logs

**Core Function:** Chronological, tamper-evident record of all user logins, scans, and supervisory reviews.

|                      |                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer navigates to Audit Trail tab; reviews event logs with timestamps, actions, and IP addresses.    |
| Input Data:          | AuditLog SQL records.                                                                                   |
| Output Produced:     | Audit table showing Action (e.g. USER_LOGIN, PRODUCT_SCANNED), Entity ID, Details, User, and Timestamp. |
| Internal Logic:      | Backend routes automatically insert AuditLog rows upon every significant mutation or scan.              |
| Operational Utility: | Ensures accountability, prevents evidence tampering, and satisfies administrative vigilance standards.  |
| Underlying Tech:     | SQLAlchemy AuditLog table, ISO timestamping.                                                            |
| Failure Handling:    | Read-only for inspectors; admin can view full history.                                                  |
| Limitations:         | Production deployment should forward logs to WORM (Write Once Read Many) cloud storage.                 |

## 18. Senior-Friendly Accessibility (Senior Mode)

**Core Function:** Enlarges typography, increases tap targets, and expands spacing for senior enforcement officers.

|                   |                                                  |
|-------------------|--------------------------------------------------|
| User Interaction: | Officer toggles 'Senior Mode' in the top navbar. |
| Input Data:       | Boolean state toggle.                            |

|                      |                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------|
| Output Produced:     | Applies .senior-mode and .large-text CSS classes across the entire DOM tree.                          |
| Internal Logic:      | CSS root font-size scaling, minimum 16px text sizes, enhanced padding on all buttons and table cells. |
| Operational Utility: | Supports senior directors and field officers who may experience vision fatigue.                       |
| Underlying Tech:     | React Context (AccessibilityContext), LocalStorage, CSS variables.                                    |
| Failure Handling:    | None.                                                                                                 |
| Limitations:         | State persists across browser reloads via LocalStorage.                                               |

19. High Contrast Visual Filter

Core Function: Enhances contrast ratios for inspection under bright sunlight or outdoor field conditions.

|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| User Interaction:    | Officer toggles 'High Contrast' in the top navbar.                                            |
| Input Data:          | Boolean state toggle.                                                                         |
| Output Produced:     | Applies .high-contrast class with sharp borders, deep black text, and high-visibility badges. |
| Internal Logic:      | CSS filter adjustments and high-contrast color token substitutions.                           |
| Operational Utility: | Ensures screen readability in outdoor wholesale grain markets and bright sunlight.            |
| Underlying Tech:     | React AccessibilityContext, CSS filters.                                                      |
| Failure Handling:    | None.                                                                                         |
| Limitations:         | Complies with WCAG 2.1 AAA contrast standards.                                                |

20. Voice & Audio Guidance (Speech Synthesis)

Core Function: Speaks inspection scores, compliance status, and action notifications aloud.

|                      |                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|
| User Interaction:    | Officer toggles 'Audio Guidance' and interacts with the application.                 |
| Input Data:          | Text strings from inspection results and status updates.                             |
| Output Produced:     | Spoken audio narration via browser speech engine.                                    |
| Internal Logic:      | Calls browser window.speechSynthesis.speak() with SpeechSynthesisUtterance.          |
| Operational Utility: | Provides hands-free audio confirmation during field inspections.                     |
| Underlying Tech:     | HTML5 Web Speech API (SpeechSynthesis).                                              |
| Failure Handling:    | Silent fallback if browser speech synthesis is unsupported or muted.                 |
| Limitations:         | Voice pitch and rate are normalized to 0.95x for clear Indian English pronunciation. |

21. SIH Instant Demo Scenario Workflows

Core Function: Pre-configured statutory packaging scenarios for instantaneous evaluator demonstration.

|                      |                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer clicks Sample 1 (Atta), Sample 2 (Oil), or Sample 3 (Namkeen).                              |
| Input Data:          | Curated image paths and statutory expected outcomes.                                                |
| Output Produced:     | Executes full live pipeline on the sample, generating real scores, bounding boxes, and PDF reports. |
| Internal Logic:      | Copies sample image from backend/samples to backend/uploads and runs the complete AI pipeline.      |
| Operational Utility: | Enables flawless, reproducible 30-second live demonstrations during hackathon judging.              |
| Underlying Tech:     | FastAPI endpoint /api/scan/demo-samples, Shutil file copy.                                          |
| Failure Handling:    | None (fully self-contained).                                                                        |
| Limitations:         | Provides exact benchmarks (Atta ~96/100, Oil ~68/100, Namkeen ~32/100).                             |

22. Role-Based Access Control (RBAC)

Core Function: Enforces hierarchical permissions across ADMIN, SUPERVISOR, and INSPECTOR roles.

|                      |                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer logs in; UI adapts available actions based on user role.                                  |
| Input Data:          | User role claim in JWT token.                                                                     |
| Output Produced:     | Restricted UI elements (e.g., supervisor review notes, audit log clearance).                      |
| Internal Logic:      | FastAPI require_roles() dependency checks user role against required permission list.             |
| Operational Utility: | Maintains administrative separation of duties between field inspectors and approving supervisors. |
| Underlying Tech:     | FastAPI Dependency Injection, JWT Role Claims.                                                    |
| Failure Handling:    | Unauthorized role actions return 403 Forbidden.                                                   |
| Limitations:         | Enforces 3 distinct tiers: ADMIN (Apex), SUPERVISOR (Review), INSPECTOR (Field).                  |

23. 1-Click Role Switcher

**Core Function:** Instantaneous role switching toolbar for evaluator demonstrations.

|                      |                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------|
| User Interaction:    | Officer clicks 'Switch to Supervisor', 'Switch to Admin', or 'Switch to Inspector' in navbar.                        |
| Input Data:          | Target role string.                                                                                                  |
| Output Produced:     | Automatically authenticates as the designated official user and updates UI state.                                    |
| Internal Logic:      | Calls switchDemoRole() in AuthContext, authenticating as Soutik (Admin), Sayantan (Supervisor), or Jiya (Inspector). |
| Operational Utility: | Allows judges to test multi-role workflows in seconds without manual re-typing.                                      |
| Underlying Tech:     | React Context, FastAPI OAuth2 Login.                                                                                 |
| Failure Handling:    | None.                                                                                                                |
| Limitations:         | Demo convenience feature; disabled in production builds.                                                             |

24. Remembered Login & Credential Persistence

**Core Function:** Persists authentication state and remembered email across sessions.

|                      |                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------|
| User Interaction:    | Officer checks 'Remember Me' on the login screen.                                         |
| Input Data:          | User email string.                                                                        |
| Output Produced:     | Pre-populates email field upon subsequent visits and retains active JWT token in storage. |
| Internal Logic:      | Stores token and user payload in browser localStorage; validates on app initialization.   |
| Operational Utility: | Prevents officer frustration from repeated logins during field audits.                    |
| Underlying Tech:     | HTML5 LocalStorage API.                                                                   |
| Failure Handling:    | Clearing browser cache removes stored credentials safely.                                 |
| Limitations:         | Token expires automatically after 24 hours per security policy.                           |

## PART 8 — AI / COMPUTER VISION / OCR DEEP DIVE

*Complete technical analysis of the computer vision and optical recognition pipeline.*

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A critical distinction in PRAMAN AI that impresses technical judges is the strict architectural separation of responsibilities across the pipeline:

### 1. Optical Character Recognition (OCR)

Perceptual pattern recognition. Converts 2D pixel grids into discrete ASCII/Unicode characters with 2D bounding box coordinates and statistical character confidence. Handled by **Tesseract OCR 5.4.0**.

### 2. Information Extraction (NLP/Regex)

Structural text parsing. Groups tokens into semantic Legal Metrology declaration fields (e.g. Net Qty = 5.0 kg, MRP = ₹230.00). Handled by **DeclarationExtractor** using deterministic linguistic patterns.

### 3. Rule Validation (Symbolic Metrology)

Deterministic legal logic. Evaluates extracted values against statutory gazettes (e.g. verifying SI unit metric compliance, tax wording, and mathematical USP consistency). Handled by **ComplianceRuleEngine**.

### 4. AI-Assisted Decision Support

Synthesis and scoring. Computes explainable 0–100 compliance scores, identifies critical infractions, and formulates statutory show-cause recommendations under Section 36(1).

## Computer Vision Edge-Case Handling in PRAMAN AI:

- **Uneven Lighting & Specular Glare:** Packaging films reflect light. PRAMAN AI applies **CLAHE (Contrast Limited Adaptive Histogram Equalization)** with `tileGridSize (8,8)` and `clipLimit 2.0` to normalize localized illumination without blowing out highlights.
- **Packaging Texture Noise:** Grainy paper, burlap bags, or textured plastic can introduce OCR noise. PRAMAN AI uses a **Bilateral Filter** (`d=9, sigmaColor=75, sigmaSpace=75`) which smooths surface texture while preserving sharp character edges.
- **Rotated & Skewed Packaging:** Handheld photos are frequently tilted. PRAMAN AI applies **Canny Edge Detection and Probabilistic Hough Transform** (`cv2.HoughLinesP`) to detect dominant line angles, computing the median tilt angle and rotating the affine matrix back to horizontal alignment.
- **Low Contrast Micro-Lettering:** Manufacturers sometimes print declarations in faint grey text on white backgrounds. PRAMAN AI computes the standard deviation of grayscale pixels; if contrast STD is below 35, it flags Rule 10 (Principal Display Panel Legibility) for manual officer verification.
- **Curved Bottles & Cans:** Cylindrical distortion causes text compression at boundaries. In PRAMAN AI, line-level aggregation merges adjacent bounding boxes, ensuring that even wrapped words are captured into cohesive lines.

PART 9 — LEGAL METROLOGY / COMPLIANCE LOGIC (12 RULES)

Statutory rules derived from 40 Official Gazettes and evaluated deterministically.

PRAMAN AI implements 12 distinct statutory rule groups derived directly from the Legal Metrology Act, 2009, the Legal Metrology (Packaged Commodities) Rules, 2011, and 40 indexed official gazette notifications:

[LM-PCR-01] Manufacturer / Packer / Importer Name & Complete Address (Rule 6(1)(a) & GSR 226(E))

|                     |                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| Statutory Basis:    | Rule 6(1)(a) & GSR 226(E)   Gazette: 8(xii)_0_1732871346.pdf   Severity: <b>HIGH</b>                           |
| What is Checked:    | Business entity name + complete physical address with locality, state, and 6-digit PIN code.                   |
| System Looks For:   | Explicit entity identifier (e.g. 'Manufactured by', 'Packed by', 'Mfg & Pkd by') followed by complete address. |
| Compliant Criteria: | Identifiable legal entity name AND valid 6-digit PIN code or clear locality address.                           |
| Violation Criteria: | Total absence of manufacturer name or incomplete address lacking city/state/PIN code (warning).                |
| Visual Evidence:    | Green bounding box over address block; Warning card if address lacks PIN code.                                 |

[LM-PCR-02] Common or Generic Commodity Name (Rule 6(1)(b))

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Statutory Basis:    | Rule 6(1)(b)   Gazette: 8(xii)_0_1732871346.pdf   Severity: <b>HIGH</b>                                   |
| What is Checked:    | Recognized common or generic commodity name (e.g. 'Wheat Flour / Atta', 'Sunflower Oil', 'Cotton Shirt'). |
| System Looks For:   | Explicit commodity title rather than exclusively an uninformative trademark brand name.                   |
| Compliant Criteria: | Generic commodity name clearly detected on principal packaging surface.                                   |
| Violation Criteria: | Absence of generic commodity name; package contains only trademark brand without stating contents.        |
| Visual Evidence:    | Bounding box highlighting detected commodity title card.                                                  |

[LM-PCR-03] Metric Net Quantity & Prohibited Qualifiers (Rule 6(1)(c), Rule 11, 12, 13)

|                     |                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Statutory Basis:    | Rule 6(1)(c), Rule 11, 12, 13   Gazette: 8(xii)_0_1732871346.pdf & 2023.12.29 SOP Oil   Severity: <b>CRITICAL</b>                                        |
| What is Checked:    | Net quantity expressed strictly in standard SI metric units (g, kg, ml, L, m, cm, mm, N).                                                                |
| System Looks For:   | Numeric magnitude followed by recognized SI unit symbol. Prohibits qualifiers like 'Jumbo', 'Family Pack', 'Super Saver' without standard metric weight. |
| Compliant Criteria: | Standard metric quantity declared (e.g. '5.0 kg', '1.0 L', '500 g').                                                                                     |
| Violation Criteria: | Non-metric units (lbs, oz), missing quantity, or deceptive non-standard expressions ('Jumbo Pack').                                                      |
| Visual Evidence:    | Red bounding box on prohibited qualifier or Green box on valid metric declaration.                                                                       |

[LM-PCR-04] Month and Year of Manufacture / Packing (Rule 6(1)(d))

|                     |                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| Statutory Basis:    | Rule 6(1)(d)   Gazette: 8(xii)_0_1732871346.pdf   Severity: <b>HIGH</b>                                       |
| What is Checked:    | Month and Year in which commodity was manufactured, packed, or imported.                                      |
| System Looks For:   | Valid date format ('MM/YYYY', 'Month YYYY', 'DD/MM/YYYY') preceded by 'Mfg Date', 'Pkd Date', or 'Packed on'. |
| Compliant Criteria: | Legible Month and Year declaration detected.                                                                  |
| Violation Criteria: | Complete omission of manufacturing or packing date.                                                           |
| Visual Evidence:    | Bounding box overlay on manufacturing date cluster.                                                           |

[LM-PCR-05] Maximum Retail Price (MRP) & Inclusive of All Taxes (Rule 6(1)(e) & Rule 18)

|                     |                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| Statutory Basis:    | Rule 6(1)(e) & Rule 18   Gazette: 8(xii)_0_1732871346.pdf & 230946_1732871433.pdf   Severity: <b>CRITICAL</b>            |
| What is Checked:    | Retail sale price in Indian Rupees (₹ or Rs.) with mandatory tax clause.                                                 |
| System Looks For:   | Must declare 'Maximum Retail Price ₹... (inclusive of all taxes)' or 'MRP ₹... incl. of all taxes'. Dual MRP is illegal. |
| Compliant Criteria: | MRP declared with explicit Indian currency symbol AND statutory 'inclusive of all taxes' text.                           |
| Violation Criteria: | MRP missing, currency symbol missing, or statutory 'inclusive of all taxes' clause omitted (High violation).             |
| Visual Evidence:    | Red bounding box if tax clause is missing; Green box if fully compliant.                                                 |

**[LM-PCR-06] Unit Sale Price (USP) Declaration & Math Consistency (Rule 6(1)(11) & GSR 226(E))**

|                            |                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Statutory Basis:</b>    | Rule 6(1)(11) & GSR 226(E)   Gazette: GSR226_1732871458.pdf & Amendment 2023   Severity: <b>HIGH</b>         |
| <b>What is Checked:</b>    | Unit sale price rounded to 2 decimals (per g if <1kg, per kg if >=1kg, per ml if <1L, per L if >=1L, per N). |
| <b>System Looks For:</b>   | Declared USP string ('■ X / kg') + Mathematical verification: USP must equal MRP divided by Net Quantity.    |
| <b>Compliant Criteria:</b> | USP declared in correct metric unit AND mathematically matches MRP / Quantity (within ■0.05 tolerance).      |
| <b>Violation Criteria:</b> | USP missing entirely (on packages > 1 unit) OR declared USP mathematically contradicts MRP / Quantity.       |
| <b>Visual Evidence:</b>    | Green box on valid USP; Amber warning card detailing mathematical discrepancy if mismatched.                 |

**[LM-PCR-07] Consumer Care Redressal Details (Rule 6(1)(f) & Rule 2(aa))**

|                            |                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Statutory Basis:</b>    | Rule 6(1)(f) & Rule 2(aa)   Gazette: 8(xii)_0_1732871346.pdf & 2022 Garment Amendment   Severity: <b>HIGH</b> |
| <b>What is Checked:</b>    | Name, address, telephone / toll-free helpline number, and email address of consumer redressal officer.        |
| <b>System Looks For:</b>   | At least two active direct contact modes: valid Email address AND Telephone / Helpline number.                |
| <b>Compliant Criteria:</b> | Both valid email address AND telephone helpline detected.                                                     |
| <b>Violation Criteria:</b> | Complete omission of consumer care details, or partial disclosure (only phone or only email - warning).       |
| <b>Visual Evidence:</b>    | Bounding box on consumer care panel; Warning card if only one contact mode is present.                        |

**[LM-PCR-08] Country of Origin (COO) Declaration (Rule 6(1)(da) & Rule 6(4))**

|                            |                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Statutory Basis:</b>    | Rule 6(1)(da) & Rule 6(4)   Gazette: 2026.02.13 COO Filter & PCR 3rd 2026   Severity: <b>HIGH</b>     |
| <b>What is Checked:</b>    | Explicit Country of Origin for all imported goods and e-commerce pre-packaged commodities.            |
| <b>System Looks For:</b>   | 'Country of Origin: ', 'Made in ', or 'Manufactured in '.                                             |
| <b>Compliant Criteria:</b> | Explicit origin country declared, or domestic origin verified via manufacturer address stating India. |
| <b>Violation Criteria:</b> | Absence of Country of Origin declaration on imported or e-commerce packaging.                         |
| <b>Visual Evidence:</b>    | Bounding box on 'Made in India' / Origin declaration.                                                 |

**[LM-PCR-09] Garment & Hosiery Standard Size & Piece Count (3rd Amendment 2022 / GSR 858(E))**

|                            |                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Statutory Basis:</b>    | 3rd Amendment 2022 / GSR 858(E)   Gazette: 2022 3rd amendment in PCR Garments   Severity: <b>MEDIUM</b>       |
| <b>What is Checked:</b>    | Standard international/Indian size code (S, M, L, XL, XXL) or dimensional measurements (cm/in) + piece count. |
| <b>System Looks For:</b>   | Apparel category packages must declare size code and count ('1 N' / '1 Piece').                               |
| <b>Compliant Criteria:</b> | Standard size code or dimensional measurement detected on garment packaging.                                  |
| <b>Violation Criteria:</b> | Garment package lacking standard size designation or piece count.                                             |
| <b>Visual Evidence:</b>    | Bounding box on apparel size label.                                                                           |

**[LM-PCR-10] Principal Display Panel Legibility & Contrast (Rule 7 & Rule 9)**

|                            |                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------|
| <b>Statutory Basis:</b>    | Rule 7 & Rule 9   Gazette: 8(xii)_0_1732871346.pdf & 267107_1761404707.pdf   Severity: <b>MEDIUM</b> |
| <b>What is Checked:</b>    | Mandatory declarations must appear on PDP with minimum numeral height and sharp contrast.            |
| <b>System Looks For:</b>   | OpenCV image contrast standard deviation and numeral bounding box height analysis.                   |
| <b>Compliant Criteria:</b> | High contrast text (Contrast STD >= 40, brightness 60–210).                                          |
| <b>Violation Criteria:</b> | Low contrast (STD < 25) or poor lighting triggering 'MANUAL_VERIFICATION_REQUIRED'.                  |
| <b>Visual Evidence:</b>    | Contrast and lighting telemetry card in inspection report.                                           |

**[LM-PCR-11] QR Code & Digital Declarations Proviso (Rule 6(1)(a) proviso & QR Amendment)**

|                            |                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Statutory Basis:</b>    | Rule 6(1)(a) proviso & QR Amendment   Gazette: Notification - Legal Metrology (QR Code)_1732871487.pdf   Severity: <b>MEDIUM</b> |
| <b>What is Checked:</b>    | QR code presence does not exempt physical label from declaring Manufacturer, MRP, Net Qty, and Helpline.                         |
| <b>System Looks For:</b>   | Detects QR code references; verifies that physical mandatory declarations are preserved.                                         |
| <b>Compliant Criteria:</b> | QR code present while all mandatory physical declarations remain on the package surface.                                         |
| <b>Violation Criteria:</b> | QR code present, but manufacturer omits physical MRP or Net Qty claiming 'scan for details' (Illegal).                           |
| <b>Visual Evidence:</b>    | Visual bounding box on QR code reference with proviso verification badge.                                                        |

[LM-PCR-12] Rule 26 Exemption Verification & Pan Masala Restriction (Rule 26 & 2025 Amendment)

|                     |                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------|
| Statutory Basis:    | Rule 26 & 2025 Amendment   Gazette: 2nd PCR Pan Masala_1764736734.pdf   Severity: <b>LOW</b>             |
| What is Checked:    | Packages <= 10g/10ml exempt from certain declarations EXCEPT Pan Masala / Tobacco where all rules apply. |
| System Looks For:   | Evaluates package category and declared net quantity against statutory exemption thresholds.             |
| Compliant Criteria: | Valid exemption for qualified micro-packages or bulk containers (>25kg).                                 |
| Violation Criteria: | Pan Masala package omitting declarations claiming small package exemption (Illegal).                     |
| Visual Evidence:    | Exemption audit badge in inspection results.                                                             |

## PART 10 — NUTRITIONAL VALUES & HEALTH CLASSIFICATION

Nutritional facts extraction, FSSAI/ICMR threshold evaluation, and health indicators.

**CRITICAL STATUTORY DISTINCTION: LEGAL COMPLIANCE ≠ HEALTH ASSESSMENT**

A pre-packaged food product can be **100% Legally Compliant** under the Legal Metrology Act, 2009 while simultaneously possessing a **'LESS HEALTHY'** nutritional profile (e.g., deep-fried potato chips with complete mandatory declarations). Conversely, a nutritious whole-food product can be **'NON-COMPLIANT'** under Legal Metrology if it omits statutory tax wording. PRAMAN AI strictly decouples these two analytical domains.

| Nutrient Field       | Standard Benchmark / Threshold                  | Evaluation & Health Logic                                                                               |
|----------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Energy / Calories    | Declared in kcal / kJ per 100g or 100ml         | Evaluates overall caloric density. High energy (>500 kcal/100g) flagged for snack categories.           |
| Protein              | >= 10.0g/100g is Rich Source; >= 5.0g is Source | Positive nutritional contributor. Increases health score.                                               |
| Dietary Fiber        | >= 6.0g/100g is High Fiber; >= 3.0g is Source   | Strong positive factor for digestive health and metabolic balance.                                      |
| Sodium / Salt        | > 600 mg/100g is HIGH; < 120 mg/100g is LOW     | Primary nutrient of public health concern. Elevated sodium triggers 'LESS HEALTHY' risk classification. |
| Saturated Fat        | > 5.0 g/100g is HIGH; < 1.5 g/100g is LOW       | Cardiovascular risk factor. Pure cooking oils/fats are handled contextually as culinary media.          |
| Total & Added Sugars | > 22.5 g/100g is HIGH; < 5.0 g/100g is LOW      | High sugar density triggers public health risk flag.                                                    |
| Trans Fatty Acids    | Must be 0.0 g/100g; > 0.2g triggers risk flag   | Zero trans fat is positively recognized per FSSAI regulations.                                          |

## PART 11 — VISUAL EVIDENCE & OVERLAY PIPELINE

How PRAMAN AI links raw packaging pixels directly to legal infractions.

In legal metrology enforcement, an unbacked accusation of non-compliance is easily dismissed in court. PRAMAN AI builds an unbroken **Chain of Visual Evidence**:

- 1. Image Pixel Matrix:** Raw uploaded package photograph stored with SHA-256 integrity hash.
- 2. OCR Coordinate Mapping:** Tesseract extracts word-level bounding box coordinates (x, y, w, h).
- 3. Declaration Field Association:** Regex parser identifies the statutory declaration and binds it to the exact line coordinates.
- 4. Deterministic Rule Evaluation:** ComplianceRuleEngine determines if the declared value satisfies statutory criteria.
- 5. Color-Coded Overlay Rendering:** OpenCV renders a high-visibility rectangle (Green for PASS, Amber for WARNING, Red for VIOLATION) and prints the statutory Rule ID badge.
- 6. Evidence Crop Extraction:** OpenCV slices the bounding box region with a 20-pixel safety margin and saves a standalone high-resolution evidence snippet JPEG.
- 7. Court-Ready Report Attachment:** The evidence crop URL and spatial coordinates are permanently embedded into the official PDF/Word inspection report.

PART 12 — REPORT GENERATION ENGINE (PDF & DOCX)

Official statutory notice generation with ReportLab and python-docx.

PRAMAN AI includes dual statutory report generators configured to match standard Government of India administrative and judicial formats:

| Report Feature / Parameter | Official PDF Notice (ReportLab)                                                                    | Editable Word Notice (python-docx)                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Target Use-Case            | Court-admissible statutory notice, immutable PDF archive, supervisor digital signature.            | Editable working draft for legal officers to append court case numbers or custom notes.         |
| Document Layout            | Formal Government of India banner, Ministry header, statutory metadata grid, 4 structured tables.  | Standard OpenXML document with formatted headings, metadata tables, and numbered finding lists. |
| Declarations Audit Table   | Includes all 9 mandatory declarations with FOUND/MISSING status badges and confidence percentages. | Complete 4-column Word table detailing field names, values, detection status, and confidence.   |
| Nutritional Information    | Dedicated nutritional breakdown table + AI health classification summary and disclaimer.           | Summary paragraph detailing detected nutritional status and public health indicator.            |
| Violations & Findings      | Color-coded status badges (PASS / WARNING / VIOLATION), source gazette citations, and findings.    | Bulleted rule evaluations with legal reference citations and specific explanatory findings.     |
| Endorsement Block          | Dual official sign-off boxes for Inspecting Official and Assistant Controller / Controller.        | Official endorsement signature block at document terminus.                                      |
| Generation Speed           | Sub-second (< 200 ms) via ReportLab Platypus engine.                                               | Instantaneous (< 100 ms) via python-docx engine.                                                |

PART 13 — USER ROLES & PERMISSIONS (RBAC)

Role-Based Access Control matrix verified in the backend and frontend.

PRAMAN AI implements 3 distinct operational roles mapped to real-world Legal Metrology department hierarchies:

| Role Name  | Authorized Personnel & Badge                    | Department / Unit                                         | Access Privileges & Restrictions                                                                                                                                          |
|------------|-------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADMIN      | Soutik (LM-ADM-001)<br>Rimi (LM-ADM-004)        | Directorate of Legal Metrology, Central HQ & Apex Command | Full system access: scan packaging, view all inspections, manage users, clear audit logs, view analytics, download all reports.                                           |
| SUPERVISOR | Sayantan (LM-SUP-102)<br>Debopriya (LM-SUP-105) | Regional Standards & Legal Enforcement Zones I & II       | Supervisory access: perform scans, review inspections, append official supervisor notes, approve or flag inspections, export notices. Cannot clear audit logs.            |
| INSPECTOR  | Jiya (LM-INS-203)<br>Arkadip (LM-INS-206)       | Field Inspection & Metrological Vigilance Unit            | Field access: scan packaging, view own and regional inspection history, download inspection reports, search products and rules. Cannot approve inspections or clear logs. |

## PART 14 — COMPLETE DATA FLOW & SCHEMAS

Conceptual data objects and state lifecycle across the inspection pipeline.

The lifecycle of data in PRAMAN AI progresses through 7 well-defined structural schemas:

**1. Ingestion Payload:**

Multipart image file or sample string → stored on disk at backend/uploads/scan\_UUID.jpg.

**2. Image Metrics Object:**

{mean\_brightness: float, contrast\_std: float, width: int, height: int, is\_low\_contrast: bool, quality\_rating: 'GOOD'|'FAIR'|'LOW'}.

**3. OCR Raw & Line Tokens:**

{raw\_text: str, average\_confidence: float, lines: [{text, x, y, w, h, confidence}], blocks: [...]}.

**4. Extracted Declarations Map:**

{commodity\_name: {value, found, confidence, bbox}, manufacturer\_details: {value, found, has\_complete\_address, bbox}, net\_quantity: {value, numeric\_value, unit, is\_standard\_unit, bbox}, mrp: {value, numeric\_value, includes\_taxes\_text, bbox}, unit\_sale\_price: {value, numeric\_value, unit, bbox}, consumer\_care: {value, email, phone, has\_email, has\_phone, bbox}, country\_of\_origin: {value, found, bbox}, garment\_size: {...}, qr\_code\_present: bool}.

**5. Nutritional Profile Object:**

{energy: {value, numeric\_value, unit, per}, protein: {...}, carbohydrates: {...}, sugars: {...}, added\_sugars: {...}, fat: {...}, saturated\_fat: {...}, trans\_fat: {...}, sodium: {...}, fiber: {...}, serving\_size: {...}}.

**6. Health Assessment Object:**

{classification: 'HEALTHIER'|'MODERATE'|'HIGH / LESS HEALTHY'|'INSUFFICIENT DATA', score: int, positive\_factors: [str], risk\_factors: [str], disclaimer: str}.

**7. Compliance Results & Score Record:**

{overall\_score: int (0-100), decision: 'COMPLIANT'|'NON-COMPLIANT'|'PENDING REVIEW', decision\_summary: str, recommended\_action: str, categories: [{name, score, max\_score, percentage}], results: [{rule\_id, rule\_name, category, status, severity, detected\_value, expected\_condition, explanation, source\_document, source\_section, bbox, crop\_url}], violations: [...]}.

## PART 15 — TECHNOLOGY STACK & TOOL JUSTIFICATIONS

Exhaustive audit of all technologies, libraries, and frameworks utilized.

| Layer / Technology   | Version / Package             | Role in Project                  | Engineering Justification                                                                             |
|----------------------|-------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Backend Core         | Python 3.12 + FastAPI         | Asynchronous REST API Gateway    | High performance async IO, automatic OpenAPI Swagger docs, native Pydantic validation.                |
| Computer Vision      | OpenCV 4.10 (cv2) + NumPy     | Image Preprocessing & Overlays   | Industry standard C++ optimized image processing for CLAHE, denoising, and bounding box drawing.      |
| OCR Engine           | Tesseract 5.4.0 + pytesseract | Optical Character Recognition    | Open-source, local, zero-cloud dependency, word-level bounding box and confidence output.             |
| Symbolic Rule Engine | Custom Deterministic Python   | Legal Metrology Compliance Logic | Zero hallucination, 100% explainable, mathematically verifiable, statutory gazette grounded.          |
| PDF Report Engine    | ReportLab 4.5.1 Platypus      | Official PDF Notice Generator    | Precision typographic control, court-admissible formatting, multi-page flowable architecture.         |
| Word Report Engine   | python-docx 1.1.2             | Editable DOCX Report Generator   | Generates standards-compliant OpenXML documents editable in Microsoft Word and LibreOffice.           |
| Frontend Framework   | React 18 + Vite 5             | Single Page Application (SPA)    | Instant Hot Module Replacement, modular component architecture, smooth client-side navigation.        |
| Styling System       | Tailwind CSS 3.4              | UI Design & Senior Accessibility | Utility-first design tokens, responsive layouts, rapid prototyping of senior-friendly visual modes.   |
| Database & ORM       | SQLAlchemy 2.0 + SQLite       | Relational Data Persistence      | Zero-configuration local deployment, seamless migration path to PostgreSQL for enterprise production. |

|                 |                           |                       |                                                                                      |
|-----------------|---------------------------|-----------------------|--------------------------------------------------------------------------------------|
| Auth & Security | PyJWT + Passlib + hashlib | Authentication & RBAC | Stateless JSON Web Tokens with salted SHA-256 password hashing and role enforcement. |
|-----------------|---------------------------|-----------------------|--------------------------------------------------------------------------------------|

## PART 16 — TECHNICAL DESIGN DECISIONS & TRADE-OFFS

Key architectural choices, rationales, benefits, trade-offs, and discarded alternatives.

### Decision 1: Deterministic Rule Engine vs Generative LLM

- **Rationale:** Why Chosen: In statutory legal enforcement, decisions must be 100% reproducible, explainable, and legally defensible in court. LLMs suffer from non-deterministic hallucinations, token limits, and high API costs.
- **Primary Benefit:** Benefit: Guaranteed correctness, zero hallucination, sub-millisecond execution, mathematically verifiable USP checks.
- **Architectural Trade-off:** Trade-off: Requires manual rule definition derived from gazette amendments rather than zero-shot prompt parsing.
- **Alternative Considered:** Discarded Alternative: Multi-modal LLM (GPT-4o / Claude 3.5 Sonnet) prompt-based compliance checking.

### Decision 2: Local OpenCV + Tesseract Pipeline vs Cloud Vision APIs

- **Rationale:** Why Chosen: Enforcement officers frequently operate in remote wholesale markets, agricultural mandis, and border checkpoints with intermittent internet connectivity.
- **Primary Benefit:** Benefit: 100% offline edge capability, zero recurring per-image API costs, absolute data privacy for confidential packaging audits.
- **Architectural Trade-off:** Trade-off: Tesseract requires dedicated pre-processing (CLAHE, deskewing) to match cloud OCR accuracy on noisy images.
- **Alternative Considered:** Discarded Alternative: Google Cloud Vision API / AWS Textract.

### Decision 3: Dual Report Generation (ReportLab PDF + python-docx DOCX)

- **Rationale:** Why Chosen: Government legal proceedings require both an immutable official document (PDF) and an editable working notice (Word) for legal clerks to append court case numbers.
- **Primary Benefit:** Benefit: Satisfies both formal court archival requirements and internal legal drafting workflows simultaneously.
- **Architectural Trade-off:** Trade-off: Requires maintaining two separate template generation codebases.
- **Alternative Considered:** Discarded Alternative: Single HTML-to-PDF headless browser conversion (Puppeteer/Weasyprint).

### Decision 4: Spatial Line Grouping Parser vs Raw Text Regex

- **Rationale:** Why Chosen: Packaging text is arranged in distinct visual clusters (e.g. price block, manufacturer block). Searching raw unstructured strings can confuse adjacent labels.
- **Primary Benefit:** Benefit: Binds bounding box coordinates directly to extracted declaration tokens, enabling precision visual evidence card cropping.
- **Architectural Trade-off:** Trade-off: Slightly higher memory consumption during line aggregation.
- **Alternative Considered:** Discarded Alternative: Flat string regex over concatenated OCR text.

## PART 17 — BRUTALLY HONEST LIMITATIONS

Realistic appraisal of current prototype boundaries and engineering remediation pathways.

True engineering maturity requires acknowledging exact system boundaries and failure modes. PRAMAN AI's known prototype limitations include:

| Prototype Limitation           | Underlying Technical Cause                                                                                | Operational Impact                                                                             | Production Roadmap Fix                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Single-Surface Image Input     | Current pipeline processes one 2D image per scan. Large boxes have declarations spread across 4 sides.    | Inspector must scan front panel first, then scan back/side panel separately.                   | Multi-image aggregation session linking front, back, and side surfaces into one inspection audit ID. |
| Highly Stylized Artistic Fonts | Tesseract is trained on standard print typography. Novel decorative brand fonts can cause misreads.       | Brand name may be missed; generic commodity name parser takes precedence.                      | Fine-tune custom lightweight OCR model (CRNN / PaddleOCR) on Indian retail packaging dataset.        |
| Severe Cylindrical Distortion  | Labels on small bottles/cans curve sharply at edges, compressing character aspect ratios.                 | Text at extreme periphery may yield lower confidence scores.                                   | 3D cylindrical unrolling mesh transform in OpenCV preprocessing before OCR execution.                |
| Regional Dialect Packaging     | English dataset is primary. Packaging in pure Tamil, Bengali, or Gujarati requires Indic OCR models.      | Non-English text triggers manual officer verification warning.                                 | Integrate Bhashini / Indic-Tesseract language packs for all 22 official Indian languages.            |
| SQLite In-Memory Concurrency   | SQLite database is optimized for local single-file storage; concurrent write locks occur under high load. | Sufficient for prototype; limits simultaneous multi-user writes in large regional deployments. | Migrate to PostgreSQL with connection pooling (PgBouncer) for national-scale deployment.             |

## PART 18 — SECURITY & AUDIT ARCHITECTURE

*Current cryptographic posture and production enterprise security roadmap.*

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Security in PRAMAN AI is architected around data integrity, evidence preservation, and role authorization:

- **Current Authentication Implementation:** Uses stateless JWT tokens signed with HMAC-SHA256. Passwords stored using salted SHA-256 hashes. Passwords never transmitted in clear text. Tokens expire after 24 hours.
- **Role-Based Access Control (RBAC):** FastAPI dependency injection enforces strict role checks (ADMIN, SUPERVISOR, INSPECTOR). Unauthorized route access returns HTTP 403 Forbidden.
- **Immutable Audit Logging:** Every user authentication, packaging scan, report download, and supervisory status change is recorded in the SQL AuditLog table with username, action, entity ID, details, IP address, and UTC timestamp.
- **Evidence Image Integrity:** Uploaded packaging images and cropped evidence snippets are given UUID4 filenames, preventing directory traversal attacks or unauthorized file overwrite.
- **Production Security Roadmap:** 1. Upgrade to asymmetric RSA (RS256) JWT keys with rotation. 2. Integrate Government of India Single Sign-On (Parichay / Jan Parichay). 3. Encrypt uploaded evidence at rest using AES-256. 4. Forward audit logs to append-only WORM cloud storage with cryptographic timestamping.

## PART 19 & 20 — SCALABILITY & REAL-WORLD GOVERNMENT DEPLOYMENT

*Path from hackathon prototype to national-scale Legal Metrology enforcement.*

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To scale PRAMAN AI across 5,000+ enforcement officers across all Indian States and Union Territories:

**Cloud Microservices Architecture:** Deploy FastAPI backend as containerized microservices on Kubernetes (EKS / NIC Cloud). Distribute OCR and Computer Vision workloads across asynchronous Celery worker pools backed by Redis message queues.

**Database Scaling & Partitioning:** Migrate from SQLite to PostgreSQL with read replicas and geographic partitioning by State / Enforcement Zone. Use PgBouncer for handling 10,000+ concurrent inspector connections.

**National Centralized Rule Repository:** Maintain a cloud-synchronized Legal Metrology Rule Knowledge Base. When the Ministry issues a new gazette amendment, updating the centralized rule JSON immediately pushes new validation logic to all field inspector devices nationwide.

**Edge-AI Handheld Enforcement Devices:** Package the OpenCV and OCR pipeline into lightweight ONNX / TensorFlow Lite models running on ruggedized Android tablets for offline field audits in remote rural markets. Inspections sync automatically when connectivity is restored.

**Integration with e-Governance Systems:** Connect PRAMAN AI directly with the National Consumer Helpline (NCH), FSSAI FoSCoS portal, and State Legal Metrology Enforcement databases to automatically file formal show-cause notices and track penalty recovery.

PART 21 — EXHAUSTIVE HACKATHON EVALUATOR QUESTIONS (A–AL)

Exhaustive question-and-answer guide organized by technical and domain categories.

This section contains defense preparation questions covering every possible angle from professors, technical evaluators, industry judges, and domain experts.

[Cat A • Problem Statement] Q1: Why can't inspectors just check packaging labels manually? Is this really a major problem in India?

*Why the Evaluator Asks This:* The evaluator wants to verify the genuine social and economic impact of the project.

**Technical Answer:** India has billions of pre-packaged commodities in circulation across retail and e-commerce. Manual inspection takes 15–20 minutes per product, requires remembering over 40 complex gazette amendments, and produces handwritten memos that lack tamper-evident visual proof. PRAMAN AI reduces audit time to under 3 seconds with 100% deterministic accuracy and court-ready visual evidence.

**Key Defense Takeaway:** 15-20 min manual vs 3-sec automated; 40+ gazette amendments; court-ready visual evidence.

[Cat A • Problem Statement] Q2: What specific legal act and rules govern packaging compliance in India?

*Why the Evaluator Asks This:* Testing domain knowledge of Indian consumer protection laws.

**Technical Answer:** The primary governing statute is the Legal Metrology Act, 2009, enforced through the Legal Metrology (Packaged Commodities) Rules, 2011 (PCR 2011), along with subsequent gazette amendments such as GSR 226(E) for Unit Sale Price and the 2022/2023 Garment and QR Code notifications.

**Key Defense Takeaway:** Legal Metrology Act 2009; Packaged Commodities Rules 2011; GSR 226(E); Section 36(1).

[Cat B • Innovation] Q3: What makes PRAMAN AI innovative compared to generic OCR or ChatGPT?

*Why the Evaluator Asks This:* Testing whether you built a wrapper or a specialized engineering solution.

**Technical Answer:** Generic OCR merely extracts unformatted text without legal meaning. ChatGPT and LLMs hallucinate, produce non-deterministic outputs, and cannot be used for court prosecution. PRAMAN AI innovates by combining Computer Vision preprocessing, localized coordinate extraction, a deterministic statutory rule engine derived from 40 gazettes, explainable 0-100 scoring, and automated court-ready PDF/DOCX notice generation.

**Key Defense Takeaway:** Hybrid CV + Deterministic Rule Engine; zero hallucination; gazette traceability; court-ready exports.

[Cat C • AI & Machine Learning] Q4: Why did you use a deterministic rule engine instead of training an end-to-end deep learning classifier for compliance?

*Why the Evaluator Asks This:* Critical technical question on architecture decision-making.

**Technical Answer:** In legal enforcement, every decision must be 100% explainable, deterministic, and traceable to a specific statutory section. An end-to-end neural network is a black box that cannot mathematically prove why a product scored 68 vs 96 in court. We use AI for perception (Computer Vision / OCR) and symbolic deterministic logic for statutory evaluation.

**Key Defense Takeaway:** Symbolic AI + Neural Perception; legal explainability; court proof; zero black-box bias.

[Cat D • OCR Implementation] Q5: How does your OCR handle different font sizes and text alignments on complex packaging?

*Why the Evaluator Asks This:* Testing deep understanding of the OCR pipeline.

**Technical Answer:** We utilize Tesseract 5.4.0 with `image\_to\_data`, extracting word-level bounding boxes and confidence scores. Our engine groups adjacent words into lines based on vertical baseline coordinates and block IDs, merging multi-line address and price clusters into unified semantic tokens.

**Key Defense Takeaway:** Word-level bounding boxes; line grouping by baseline; spatial coordinate merging.

[Cat E • Computer Vision] Q6: What happens if an inspector uploads a tilted or poorly lit package photograph?

*Why the Evaluator Asks This:* Testing robustness of the image preprocessing pipeline.

**Technical Answer:** Our OpenCV preprocessor applies CLAHE to equalize localized lighting and bilateral filtering to reduce grain noise. It then runs Canny edge detection and Probabilistic Hough Transform to calculate the median tilt angle, rotating the affine matrix to deskew the label before OCR execution.

**Key Defense Takeaway:** CLAHE contrast equalization; bilateral noise reduction; Hough transform skew correction.

[Cat F • Information Extraction] Q7: How does the system distinguish between the Manufacturer address and the Consumer Care address?

*Why the Evaluator Asks This:* Testing NLP and token extraction logic.

**Technical Answer:** The DeclarationExtractor uses contextual keyword boundaries and regex lookaheads. Manufacturer parsing looks for 'Manufactured by', 'Packed by', or 'Mfg by' followed by address patterns and PIN codes, cutting off before 'Consumer Care' or 'MRP'. Consumer Care extraction specifically isolates email formats, toll-free 1800 numbers, and 'Helpline' headers.

**Key Defense Takeaway:** Contextual boundary parsing; regex lookaheads; keyword cutoff markers.

[Cat G • Rule Engine] Q8: How do you evaluate Unit Sale Price (USP) compliance under GSR 226(E)?

*Why the Evaluator Asks This:* Testing statutory rule implementation accuracy.

**Technical Answer:** Under GSR 226(E), commodities must declare USP per gram/ml (<1kg/1L) or per kg/L (>=1kg/1L). PRAMAN AI checks for the presence of the declared USP string, parses the numeric unit price, and mathematically verifies that USP equals MRP divided by Net Quantity within a 0.05 tolerance.

**Key Defense Takeaway:** Metric unit threshold check (<1kg vs >=1kg); mathematical verification (MRP / Quantity).

[Cat H • Legal Metrology Domain] Q9: What is a Section 36(1) Notice under the Legal Metrology Act, 2009?

*Why the Evaluator Asks This:* Testing statutory penalty knowledge.

**Technical Answer:** Section 36(1) prescribes penalties for manufacturing, packing, or selling non-standard pre-packaged commodities. First offenses carry fines up to 25,000; second offenses up to 50,000; subsequent offenses up to 1,00,000 or imprisonment up to one year. PRAMAN AI automatically recommends Section 36(1) notices for critical infractions.

**Key Defense Takeaway:** Section 36(1) penalty for non-standard packages; fines up to 1 Lakh and imprisonment.

[Cat I • Nutritional Assessment] Q10: Why does PRAMAN AI include nutritional analysis if Legal Metrology is about measurement?

*Why the Evaluator Asks This:* Testing project scope understanding.

**Technical Answer:** While Legal Metrology regulates quantity and consumer declarations, packaging labels increasingly display Nutritional Facts under FSSAI mandate. PRAMAN AI provides a unified inspection studio that extracts 11 nutrients and provides an informational health indicator, strictly distinguishing legal compliance from health assessments.

**Key Defense Takeaway:** Dual-purpose enforcement; FSSAI synergy; strictly decoupled from metrology compliance score.

[Cat J • System Architecture] Q11: Explain the multi-tier architecture and data flow of PRAMAN AI.

*Why the Evaluator Asks This:* Testing end-to-end architectural clarity.

**Technical Answer:** PRAMAN AI uses a decoupled 3-tier architecture: React 18 SPA frontend, FastAPI asynchronous backend with SQLAlchemy ORM, and a processing pipeline encompassing OpenCV preprocessors, Tesseract OCR, Deterministic Rule Engines, and ReportLab / python-docx report generators.

**Key Defense Takeaway:** React 18 frontend; FastAPI ASGI; OpenCV + Tesseract; Deterministic Engine; ReportLab/docx.

[Cat K • Database Design] Q12: What relational models exist in your database schema?

*Why the Evaluator Asks This:* Testing database modeling mastery.

**Technical Answer:** Our schema in `models.py` includes User (RBAC & auth), Product (commodity catalogue), Inspection (master scan record), ExtractedDeclaration (parsed fields), ComplianceResult (rule checks), Violation (actionable infractions), Report (document paths), and AuditLog (immutable event log).

**Key Defense Takeaway:** 8 core relational tables: User, Product, Inspection, Declarations, Results, Violations, Reports, AuditLog.

[Cat L • Backend Performance] Q13: How fast is the backend pipeline and what is the processing latency?

*Why the Evaluator Asks This:* Testing performance metrics.

**Technical Answer:** The full backend pipeline—from raw image ingestion, OpenCV enhancement, Tesseract OCR, declaration extraction, rule evaluation, scoring, visual evidence cropping, to database storage—executes in approximately 1.5 to 3.0 seconds on standard x86 CPU hardware.

**Key Defense Takeaway:** 1.5 to 3.0 seconds end-to-end; CPU-optimized; sub-second report generation.

[Cat M • Frontend & UX] Q14: What accessibility features are implemented for enforcement officers?

*Why the Evaluator Asks This:* Testing accessibility and UX design.

**Technical Answer:** PRAMAN AI includes Senior Mode (enlarging base font sizes and tap targets), High Contrast Mode (optimizing contrast for outdoor sunlight), and Web Speech Synthesis Audio Guidance (reading scores and alerts aloud for hands-free operation).

**Key Defense Takeaway:** Senior Mode (large text); High Contrast Mode (sunlight); Web Speech Audio Guidance.

[Cat N • Authentication & Security] Q15: How is user authentication secured in the application?

*Why the Evaluator Asks This:* Testing security implementation.

**Technical Answer:** We implement stateless JWT tokens using HMAC-SHA256. Passwords are encrypted using salted SHA-256 hashes. Frontend stores the token in LocalStorage and attaches Bearer authorization headers to API requests. 6 pre-seeded official user roles allow instant demonstration.

**Key Defense Takeaway:** JWT HS256; Salted SHA-256 hashing; 6 official seeded enforcement credentials.

[Cat O • Edge Cases] Q16: What happens if an image has multiple prices or crossed-out old MRPs?

*Why the Evaluator Asks This:* Testing robustness against packaging tampering.

**Technical Answer:** The MRP parser searches for explicit currency symbols and tax clauses. If multiple prices or smudging is detected, OCR confidence drops and Rule LM-PCR-05 triggers a VIOLATION or WARNING for dual/smudged MRP, recommending manual physical inspection under Rule 18.

**Key Defense Takeaway:** Dual MRP detection; Rule 18 violation; low confidence triggers manual verification.

[Cat P • Scalability] Q17: How would you scale this to handle 1 million scans per day across India?

*Why the Evaluator Asks This:* Testing enterprise scalability thinking.

**Technical Answer:** We would deploy the FastAPI backend on Kubernetes with auto-scaling pods, offload OCR processing to asynchronous Celery worker queues backed by Redis, migrate the database to a sharded PostgreSQL cluster with PgBouncer, and cache static assets and reports on Cloudflare CDN.

**Key Defense Takeaway:** Kubernetes auto-scaling; Celery + Redis worker queues; Sharded PostgreSQL; CDN caching.

[Cat Q • Accuracy & Confidence] Q18: How do you measure and report OCR confidence to the inspector?

*Why the Evaluator Asks This:* Testing AI observability.

**Technical Answer:** Tesseract computes individual character confidence scores (0–100). OCREngine aggregates these into average word and line confidence metrics. If average confidence falls below 50%, the UI displays a low-confidence warning badge and flags legibility under Rule 10.

**Key Defense Takeaway:** Character-level confidence aggregation; threshold <50% triggers legibility warning.

[Cat R • Existing Solutions] Q19: Why can't the government just use barcode scanners or GS1 databases?

*Why the Evaluator Asks This:* Testing comparative domain understanding.

**Technical Answer:** Barcodes only verify digital database entries registered by the manufacturer; they cannot detect physical packaging violations like smudged MRPs, missing local addresses, prohibited 'Jumbo' qualifiers, or unprinted tax clauses. PRAMAN AI audits the physical package as seen by the consumer.

**Key Defense Takeaway:** Barcodes verify registry data; PRAMAN AI audits physical packaging ground-truth.

[Cat S • Official Reporting] Q20: What makes the generated PDF report legally valuable?

*Why the Evaluator Asks This:* Testing understanding of government documentation.

**Technical Answer:** The PDF report is formatted as an official statutory notice under the Ministry of Consumer Affairs header. It includes full metadata, an itemized declarations audit, statutory rule citations from indexed gazettes, cropped visual evidence snippets, and dual officer endorsement signature blocks.

**Key Defense Takeaway:** Official Ministry header; itemized audit; gazette citations; evidence snippets; signature blocks.

## PART 22 — HARD TECHNICAL QUESTIONS & TRAP DEFENSE

Tough adversarial questions designed to catch candidates off guard.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>[Technical Defense] QTrap 1: How do you handle OCR errors causing false legal prosecutions?</b></p> <p><i>Why the Evaluator Asks This:</i> Evaluator is testing risk mitigation and legal safety checks.</p> <p><b>Technical Answer:</b> PRAMAN AI operates as an <b>Inspector Decision Support System</b> rather than an autonomous prosecuting agent. Every detected violation displays the exact OCR text, character confidence score, and cropped visual evidence card. High-severity infractions require the Designated Enforcement Officer to review the crop card before signing the official PDF notice. Furthermore, low OCR confidence (&lt;50%) automatically marks rules as 'MANUAL_VERIFICATION_REQUIRED' rather than issuing an immediate penalty.</p> <p><b>Key Defense Takeaway:</b> Human-in-the-loop verification; confidence thresholds prevent false penalties; inspector signs notice.</p> |
| <p><b>[Technical Defense] QTrap 2: What happens when two legal metrology gazettes appear to conflict?</b></p> <p><i>Why the Evaluator Asks This:</i> Testing understanding of statutory hierarchy and amendment precedence.</p> <p><b>Technical Answer:</b> Under Indian administrative law, specific subsequent amendments override general earlier rules (lex posterior derogat priori). For example, GSR 226(E) introduced Unit Sale Price as mandatory, overriding older package format norms. In PRAMAN AI, our 12 rule definitions represent the latest consolidated statutory ground truth from all 40 gazettes through 2026, with exact gazette PDF file traceability.</p> <p><b>Key Defense Takeaway:</b> Lex posterior legal principle; latest gazette supersedes older rules; 40 indexed PDFs.</p>                                                                                                         |
| <p><b>[Technical Defense] QTrap 3: How do you distinguish between absence of evidence and evidence of absence?</b></p> <p><i>Why the Evaluator Asks This:</i> Testing deep logical modeling in information extraction.</p> <p><b>Technical Answer:</b> If the OCR engine has high confidence (&gt;85%) across the entire package surface and a mandatory declaration like 'Inclusive of all taxes' is not found, the system registers <b>Evidence of Absence</b> (Statutory Violation). If overall OCR confidence is low (&lt;50%) due to blur or poor lighting, the system registers <b>Absence of Evidence</b> and flags the item for manual inspector verification under Rule 10.</p> <p><b>Key Defense Takeaway:</b> High confidence + missing token = Violation; Low confidence = Manual Review Required.</p>                                                                                                    |
| <p><b>[Technical Defense] QTrap 4: Why not just run a lightweight multimodal LLM like Moondream or LLaVA on the edge?</b></p> <p><i>Why the Evaluator Asks This:</i> Testing why you didn't adopt newer vision-language trends.</p> <p><b>Technical Answer:</b> While multimodal VLMs are impressive for general captioning, they lack deterministic legal verification. They cannot perform precise arithmetic validation (verifying USP = MRP / Quantity to 2 decimal places), their bounding box coordinates frequently drift, and they cannot produce explainable mathematical proofs needed to defend a prosecution in consumer court.</p> <p><b>Key Defense Takeaway:</b> VLMs fail precision arithmetic; bounding box drift; lack statutory explainability.</p>                                                                                                                                                |

## PART 23 — DEMO QUESTIONS & LIVE INTERRUPTION PREPARATION

Realistic questions asked by evaluators while testing the live running application.

|                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Demo Q1: Where did this extracted MRP come from on the packaging image?</b></p> <p>Point to the Visual Evidence tab or Declarations table on screen. Show how the extracted MRP (e.g. <span>■230.00</span>) matches the green bounding box on the price block of the packaging image, displaying a 95% confidence rating.</p>   |
| <p><b>Demo Q2: Why is the Sunflower Oil sample flagged as NON-COMPLIANT if it has an MRP?</b></p> <p>Explain that while MRP is declared, the mandatory statutory clause '(inclusive of all taxes)' is missing, and the statutory consumer care helpline/email was omitted under Rule 6(1)(f), reducing the score to 68/100.</p>       |
| <p><b>Demo Q3: Why does the Namkeen pack score so low (32/100)?</b></p> <p>Point out the CRITICAL VIOLATION under Rule 6(1)(c) and Rule 11: using the prohibited qualifying expression 'Jumbo Saver Pack' in place of standard SI metric units (e.g. 500g), alongside missing manufacturer address and missing Country of Origin.</p> |
| <p><b>Demo Q4: Can I edit the inspection report before issuing it to the company?</b></p> <p>Click 'Export DOCX'. Demonstrate that PRAMAN AI generates an editable Word document containing all findings, allowing officers to customize notice text, case numbers, and legal hearing dates.</p>                                      |

## PART 24 — PERFECT DEMO WALKTHROUGH SCRIPT

*Click-by-click, say-by-say presentation script for a 3-minute winning hackathon demo.*

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Follow this exact step-by-step presentation script during your evaluation demo:

### 0:00 - 0:30 | Introduction & Login

1. Display the Login Page. Click the 1-click official user card for **Soutik (Head Admin)**.
2. **What to Say:** *'Respected Evaluators, PRAMAN AI transforms Legal Metrology compliance in India from a 20-minute manual paperwork bottleneck into a 3-second automated audit. We are logging in as Authorized Enforcement Officer Soutik.'*

### 0:30 - 1:00 | Dashboard & Scan Studio Initiation

1. Show Executive Dashboard with compliance rate gauge, top violations chart, and state-wide scan metrics.
2. Click **'Start Packaging Scan'** to enter the Scan Packaging Studio.
3. **What to Say:** *'Our dashboard provides state-wide enforcement visibility. In our Scan Studio, inspectors can upload live package photos or choose statutory benchmark scenarios.'*

### 1:00 - 1:45 | Executing AI Pipeline on Sample 1 (Compliant Atta)

1. Select **Sample 1: Chakki Fresh Atta (5 kg)**.
2. Click **'Execute Legal Metrology Inspection'**. Watch the 5-stage progress indicator.
3. System redirects to Inspection Detail page showing Score: **96/100 (COMPLIANT)**.
4. **What to Say:** *'In under 2 seconds, OpenCV deskewed the image, Tesseract extracted word coordinates, and our deterministic rule engine verified all 12 gazette rules. Net quantity (5.0 kg), MRP with tax clause, and calculated USP (₹46.00/kg) all pass.'*

### 1:45 - 2:15 | Testing Non-Compliant Sample 3 (Critical Namkeen)

1. Return to Scan Studio, select **Sample 3: Desi Namkeen Pack**, and execute inspection.
2. Detail view shows Score: **32/100 (CRITICAL VIOLATION - NON-COMPLIANT)**.
3. Click **Visual Evidence** tab to show red bounding box on prohibited 'Jumbo Saver Pack'.
4. **What to Say:** *'Notice how PRAMAN AI immediately catches non-standard misleading expressions under Rule 11 and flags missing manufacturer address, generating an immediate Section 36(1) notice recommendation.'*

### 2:15 - 2:45 | Official Reports & Rule Library Traceability

1. Click **'Download Official PDF'**. Open the generated Court-Ready ReportLab PDF notice.
2. Click **'Rule Library (40 PDFs)'** in sidebar to demonstrate ground-truth gazette linkage.
3. **What to Say:** *'Every finding links directly to our index of 40 official gazette PDFs. With one click, the officer exports an official court-ready statutory notice complete with visual evidence and signature blocks.'*

### 2:45 - 3:00 | Accessibility & Wrap Up

1. Toggle **Senior Mode** and **High Contrast** in the top navbar.
2. Click Audio Guidance to play speech synthesis.
3. **What to Say:** *'With senior-friendly accessibility and immutable audit trails, PRAMAN AI is ready for deployment across India. Thank you!'*

## PART 25 & 26 — ELEVATOR PITCHES & ACADEMIC ARCHITECTURE

Concise pitch scripts and systems engineering explanation for technical professors.

### Explain It to a Professor (Systems & Academic Perspective):

'PRAMAN AI is an applied Computer Vision and Symbolic AI platform for regulatory document validation. From an engineering perspective, it solves a domain-specific Document Visual Question Answering (DocVQA) problem without the non-deterministic latency and hallucination risks of Large Language Models. The perceptual pipeline uses OpenCV adaptive histogram equalization and Hough transform deskewing to feed normalized pixel buffers into Tesseract 5.4. Spatial line aggregation reconstructs bounding box coordinate topologies. Our deterministic symbolic rule engine executes constraint-satisfaction algorithms across 12 statutory gazette rule groups, performing arithmetic verification on Unit Sale Prices and regular-expression pattern matching on statutory clauses. The output is a weighted, explainable scalar score (0–100) and an automated PDF/DOCX artifact compiled via flowable document layout engines.'

## PART 27 — QUICK REVISION CHEAT SHEET

Essential facts, statutory references, and core metrics for last-minute revision.

|  |                                                                                                                                                                                 |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Legal Metrology Act, 2009 & Packaged Commodities Rules, 2011 (40 Official Gazettes Indexed)                                                                                     |
|  | Section 36(1) of LM Act 2009 (Fines ₹25,000 to ₹1,00,000 / Imprisonment up to 1 year)                                                                                           |
|  | 1. Mfr Name & Address, 2. Generic Name, 3. Net Qty (SI units), 4. Mfg Date, 5. MRP (incl. taxes), 6. USP, 7. Consumer Care (Email & Tel), 8. Country of Origin, 9. Garment Size |
|  | Completeness (40%), Value & Metric Units (25%), Price & USP (15%), Consumer Care & COO (10%), Legibility (10%)                                                                  |
|  | Score >= 85 & 0 violations = COMPLIANT; Score < 70 OR >=1 critical violation = NON-COMPLIANT                                                                                    |
|  | Sample 1 Atta: ~96 (COMPLIANT)   Sample 2 Oil: ~68 (NON-COMPLIANT)   Sample 3 Namkeen: ~32 (CRITICAL VIOLATION)                                                                 |
|  | Soutik (Head Admin - 001), Sayantan (Supervisor - 102), Jiya (Inspector - 203), Rimi (Admin - 004), Debopriya (Supervisor - 105), Arkadip (Inspector - 206)                     |
|  | Python 3.12, FastAPI, React 18, Vite, OpenCV 4.10, Tesseract 5.4, ReportLab 4.5, python-docx, SQLite                                                                            |

## PART 28 — QUESTIONS WE MUST BE ABLE TO ANSWER (TEAM SELF-TEST)

Final readiness checklist before facing the hackathon judging panel.

- 1. Can you explain why Unit Sale Price (USP) is mandatory under GSR 226(E) and how our system checks its math?
- 2. Can you show where in the code bounding box coordinates are converted into cropped evidence JPEG snippets?
- 3. Can you explain why a high-fat snack can be legally compliant under Metrology while scoring 'LESS HEALTHY' in Nutrition?
- 4. Can you demonstrate the difference in UI capabilities between Soutik (Admin), Sayantan (Supervisor), and Jiya (Inspector)?
- 5. Can you explain how our OpenCV preprocessor uses CLAHE and Hough transform to handle uneven lighting and skew?
- 6. Can you download both the PDF and DOCX reports live during the demo without any server latency?

### FINAL PREPARATION DIRECTIVE

All team members must review this dossier thoroughly. Demonstrate confidence, emphasize deterministic legal explainability over AI hype, showcase live visual evidence crops, and highlight the 40 indexed statutory gazettes. Good luck!